ASIAN DEVELOPMENT BANK
RRP:PAK 34333

REPORT AND RECOMMENDATION

OF THE

PRESIDENT

TO THE

BOARD OF DIRECTORS

ON A

PROPOSED LOAN

AND

TECHNICAL ASSISTANCE LOAN

AND

TECHNICAL ASSISTANCE GRANT

TO THE

ISLAMIC REPUBLIC OF PAKISTAN

FOR THE

BALOCHISTAN ROAD DEVELOPMENT SECTOR PROJECT

October 2003

CURRENCY EQUIVALENTS
(as of 1 October 2003)

Currency Unit	–	Pakistan rupee (PRe/PRs)
PRe1.00	=	$0.0169
$1.00	=	PRs59.00

ABBREVIATIONS

ADB	–	Asian Development Bank
AC	–	asphalt concrete
AP	–	affected person
ATTA	–	Afghan Transit Trade Agreement
BOT	–	build-operate-transfer
BPG	–	Balochistan provincial government
CAS	–	compulsory acquisition surcharge
CDC	–	cross-border development component
CSC	–	construction supervision consultant
CWD	–	Communication and Works Department
EIA	–	environmental impact assessment
EIRR	–	economic internal rate of return
EMP	–	Environmental Management Plan
GDP	–	gross domestic product
GRC	–	grievance redress committee
HDM	–	Highway Design and Maintenance Model
HIV/AIDS	–	human immunodeficiency virus/acquired immunodeficiency syndrome
ICB	–	international competitive bidding
IEE	–	initial environmental examination
IRI	–	international roughness index
JBIC	–	Japan Bank for International Cooperation
LAA	–	Land Acquisition Act
LIBOR	–	London interbank offered rate
M&E	–	monitoring and evaluation
MOC	–	Ministry of Commerce
NGO	–	nongovernment organization
NH	–	national highway
NHA	–	National Highway Authority
NHC	–	national highway component
NHMP	–	National Highway and Motorway Police
NHSC	–	National Highway Safety Committee
PAC	–	provincial access component
PH	–	provincial highway
PIDC	–	provincial institutional development component
PIR	–	poverty impact ratio
PMU	–	Project Management Unit
PSP	–	private sector participation
RAC	–	resettlement advisory committee
RAMS	–	Road Asset Management System

RAR	–	rural access road
RESAC	–	Road Environment and Social Assessment Cell
RMF	–	road maintenance fund
RMU	–	road management unit
ROW	–	right-of-way
RP	–	resettlement plan
SBE	–	small business enterprise
SOP	–	standard operating procedure
TA	–	technical assistance
TST	–	surface treatment
UNDP	–	United Nations Development Programme
VOC	–	vehicle operating cost

NOTES

(i) The fiscal year (FY) of the Government ends on 30 June. FY before a calendar year denotes the year in which the fiscal year ends, e.g., FY2003 ends on 30 June 2003.

(ii) In this report, "$" refers to US dollars.

This report was prepared by a team consisting of H. Carlsson, team leader, H. Masood, D. Utami, N. Zhang, K. Ahmad, and A. Lee .

CONTENTS

SUPPLEMENTARY APPENDIXES (available upon request)

LOAN AND PROJECT SUMMARY

Borrower	Islamic Republic of Pakistan
Classification	Poverty intervention Thematic: Economic growth
Environment Assessment	Category B. An initial environmental examination (IEE) was undertaken, and the summary IEE is a core appendix.
Project Description	The Project will improve connectivity and allow better access in Balochistan, and build the capacity and systems for sustainable road sector development, specifically in road maintenance. The Project will also improve the efficiency of national highway (NH) 25, including one major border crossing, that passes through Balochistan, and forms part of the priority road transport corridor, that links Pakistan, Afghanistan, and Central Asia together.
Rationale	Balochistan is the largest and least developed province in Pakistan. The economic development potential in the province is constrained by its limited resource base; inadequate social and economic infrastructure; lack of connectivity between the poor rural population and market and district centers; and poor access to education and health facilities, and social services; and few employment opportunities. The onset of peace in Afghanistan and developments in Pakistan and Central Asia offer opportunities for regional cooperation. As one of the principal gateways between Pakistan and Afghanistan, Balochistan is well placed to play an increasing role in regional trade and cooperation if transport bottlenecks are removed and cross-border arrangements improved. The Asian Development Bank (ADB) has adopted a programmatic approach that sequences interventions in individual provinces over several years. This Project is the third intervention using that approach, following the road projects that were approved in 2001 and 2002 and covered the provinces of Sindh and Punjab, respectively. Balochistan provincial government (BPG) has taken some initiatives to address provincial road sector policy issues, and established a road management unit (RMU) at its Communications and Works Department (CWD). However, major challenges remain, including continued institutional changes at CWD and efforts to improve road maintenance and road safety. At the federal level, the National Highway Authority (NHA), which is responsible for national highway development and maintenance, is undergoing a policy and institutional reform program supported by ADB.

Objectives and Scope

The objectives of the Project are to reduce poverty and support pro-poor economic growth by (i) enhancing access of rural populations to economic opportunities and social services by improving the provincial road network; (ii) improving trade facilitation and efficiency of the priority road transport corridor linking Pakistan, Afghanistan and Central Asia together, which traverse Balochistan; and (iii) providing essential institutional support and capacity building at CWD.

The Project has four main components: (i) a provincial institutional development component (PIDC) supporting CWD and districts, (ii) a provincial access component (PAC) comprising rehabilitation and construction of about 1,100 kilometers (km) of provincial roads, (iii) a cross-border development component (CDC) supporting improved trade facilitation and providing a new pilot cross-border facility, and (iv) a national highway component (NHC) comprising rehabilitation of 247 km of national highways. Rehabilitation and construction of the provincial roads will follow a sector approach.

Cost Estimates

The total cost of the Project is estimated at $267.3 million equivalent. The foreign exchange cost is $162.5 million, and the local currency cost is $104.8 million equivalent.

Financing Plan

($ million)

Source	Foreign Exchange	Local Currency	Total Cost	Percent
ADB	162.5	23.2	185.7	69.5
Government	0.0	81.6	81.6	30.5
Total	**162.5**	**104.8**	**267.3**	**100.0**

ADB = Asian Development Bank.

Loan Amount and Terms

A loan of ¥20,266,370,000 ($185.7 million equivalent) from ADB's ordinary capital resources will be provided. The loan will have a 25-year term including a grace period of 5 years, an interest rate determined in accordance with ADB's LIBOR-based lending facility, a commitment charge of 0.75% per annum, a front-end fee of 0.5%, and such other terms and conditions set forth in the draft Loan and Project Agreements. The Borrower will make the loan proceeds available to NHA ($85.7 million equivalent) and BPG ($100 million equivalent).

Period of Utilization

Until 31 December 2009

Estimated Project Completion Date

30 June 2009

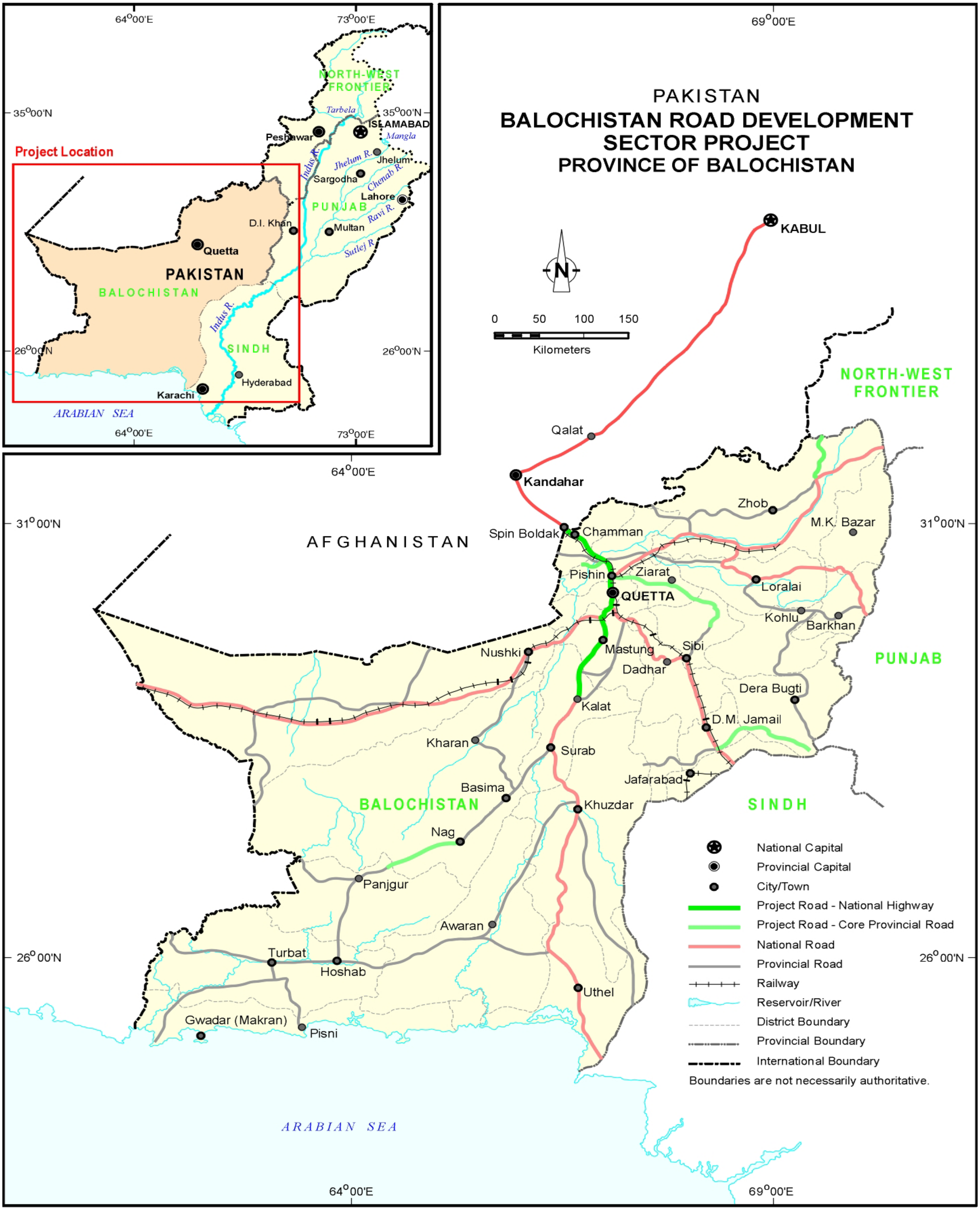
Executing Agencies	CWD Balochistan for the PIDC and the PAC, NHA for the CDC and the NHC.
Implementation Arrangements	The project components will be implemented by the established project management units (PMUs) at CWD and NHA, respectively. Each PMU will be headed by a project director. The additional chief secretary (ACS) of BPG will chair a steering group for the PIDC and the PAC. The Chairperson of NHA will chair a steering group for the CDC and the NHC.
Procurement	All procurement will be carried out in accordance with ADB's *Guidelines for Procurement*. Advance action for procurement was approved on 14 July 2003.
Consulting Services	The Project will require 250 person-months of international and 4,509 person-months of domestic consulting services for (i) institutional development with capacity building under the PIDC, (ii) detailed design and construction supervision for the civil works for the provincial roads, and (iii) construction supervision of the civil works for the national highways. The consultants will be recruited in accordance with ADB's *Guidelines on the Use of Consultants,* and other arrangements satisfactory to ADB for engaging domestic consultants.
The Technical Assistance Loan	A TA loan equivalent to $1.0 million (SDR 701,000) for Community Development and Poverty Targeting will be provided from ADB's Special Funds resources. The TA loan will have a term of 32 years, including a grace period of 8 years; an interest rate of 1% per annum during the grace period and 1.5 % per annum thereafter; and such other terms and conditions as are substantially in accordance with those set forth in the draft TA Loan Agreement.
Technical Assistance Grant	ADB will provide a TA grant in the amount of $500,000 for cross-border development. The TA will be financed on a grant basis by ADB's TA funding program.

Project Benefits and Beneficiaries

The main quantifiable benefits of the Project consist of savings in vehicle operating costs and passenger time. The economic internal rates of return (EIRR) for the selected two national highways are 33.7% and 40.4%, respectively. For the six core provincial roads, the EIRRs are 14.8–48.9%. All provincial roads were subjected to analysis based on multiple social and economic criteria. The Project will reduce poverty by (i) enabling the poor to participate in economic opportunities and enhancing their access to social services, and (ii) creating additional economic opportunities for the poor by stimulating economic growth. Project beneficiaries will comprise (i) the poor who will have increased access to economic opportunities and human development services because of improved national highways and provincial roads; (ii) the poor who live in the districts in Balochistan and were previously unable to pursue employment and other economic opportunities due to geographic isolation and long journey times; (iii) companies, employees, consumers and transport operators in Balochistan Province, Afghanistan, and Central Asia who directly or indirectly use goods and services transported on the project roads.

Risks and Assumptions

The risks to the Project are related to (i) start-up delays, (ii) commitment to institutional changes at CWD, (iii) progress on reform at NHA, (iv) adequacy of road maintenance at federal and provincial levels and (v) commitment to cross-border development. The risks will be mitigated through (i) advance actions for recruiting consultants and starting procurement activities, (ii) use of already established PMUs and appointed project directors, (iii) BPG's commitment to and agreement with ADB on institutional development and changes, (iv) policy and institutional reform measures already implemented by NHA, (v) actions initiated by NHA to improve road maintenance and increase its funding, (vi) capacity building at CWD and other support for road maintenance under the Project, and (vii) strong support by the federal Government and advance actions already under way to improve trade facilitation including transit arrangements. NHA and CWD have satisfactorily implemented past loans in the road sector.


PAKISTAN
BALOCHISTAN ROAD DEVELOPMENT SECTOR PROJECT
PROVINCE OF BALOCHISTAN
Project Location
PAKISTAN
BALOCHISTAN
NORTH-WEST FRONTIER
PUNJAB
SINDH
ISLAMABAD
Peshawar
Quetta
D.I. Khan
Multan
Lahore
Sargodha
Jhelum
Hyderabad
Karachi
Tarbela
Mangla
Indus R.
Jhelum R.
Chenab R.
Ravi R.
Sutlej R.
ARABIAN SEA
64°00'E
73°00'E
35°00'N
26°00'N
KABUL
Qalat
Kandahar
AFGHANISTAN
Spin Boldak
Chamman
Pishin
Ziarat
QUETTA
Zhob
M.K. Bazar
Loralai
Kohlu
Barkhan
Nushki
Mastung
Sibi
Dadhar
Kalat
Dera Bugti
D.M. Jamali
Kharan
Surab
Jafarabad
Basima
Khuzdar
Nag
Panjgur
Awaran
Turbat
Hoshab
Uthel
Gwadar (Makran)
Pisni
0 50 100 150
Kilometers
N
69°00'E
31°00'N
National Capital
Provincial Capital
City/Town
Project Road - National Highway
Project Road - Core Provincial Road
National Road
Provincial Road
Railway
Reservoir/River
District Boundary
Provincial Boundary
International Boundary
Boundaries are not necessarily authoritative.

03-0454a HR

I. THE PROPOSAL

1. I submit for your approval the following report and recommendation on a (i) proposed loan to the Islamic Republic of Pakistan for the Balochistan Road Development Sector Project; and (ii) a proposed technical assistance (TA) loan for Community Development and Poverty Targeting. The report also describes a proposed TA grant for Cross-Border Development, and if the Board approve the proposed loan and TA loan, I, acting under the authority delegated to me by the Board, will approve the TA grant. The project framework is in Appendix 1.

II. RATIONALE: SECTOR PERFORMANCE, PROBLEMS, AND OPPORTUNITIES

A. Performance Indicators and Analysis

2. **Road Sector.** Balochistan, which is the largest but least developed and least populated province in Pakistan, has 6.6 million people,[1] or 5% of the total population. About 76% of Balochistan's population live in rural areas. The provincial road network[2] grew from 14,682 kilometer (km) in 1993 to 23,451 km in 2000 (Appendix 2) reflecting poor connectivity in the province. The largest expansion was that of farm-to-markets roads, which increased manifold with support frp, the Asian Development Bank (ADB) and the Government of Japan. The length of the national highways remained largely constant at 2,300 km. Balochistan has still, however, the lowest density of roads among the four provinces of Pakistan. Poor connectivity and access continue to be a major problem, which particularly affect the poor, who live mostly in the rural areas. The increased demand for road transport – about 7-8% per year – and higher traffic intensity combined with insufficient maintenance have caused several parts of the road network to deteriorate. A survey of 2,500 km of provincial roads found that gravel roads or earthen tracks representing 90% of the network were in poor to very poor condition and were deteriorating rapidly owing to increased traffic, underinvestment in transport infrastructure, and insufficient maintenance.[3] The national highways in Balochistan, which are administered by the National Highway Authority (NHA), play an important role from the regional perspective as they carry transit traffic between Pakistan and Afghanistan and provide access to ports[4] in southern Pakistan through national highway (NH) 25. Several sections on NH25 are, however, constrained, too narrow with respect to the traffic carried, and in poor condition; the conditions cause high vehicle operating cost and compromise road safety.

3. **Pakistan's Regional Cooperation and Trade Potential.** The onset of peace in Afghanistan, introduction of market-oriented reforms in Pakistan, and developments in Central Asia have opened up opportunities for closer cooperation among these countries. Pakistan and its province Balochistan is geographically well placed and has the potential to play an increasing role in regional cooperation, linking landlocked Afghanistan, and Central and South Asia more closely. Pakistan provides the shortest route (about 300-500 km shorter than the second best alternative) to ports in Afghanistan and most countries in Central Asia, which are landlocked and remain dependent on road transport and transit arrangements with neighboring countries that allow the use of their port facilities. Pakistan, Afghanistan, and the countries in Central Asia

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1 From 1981 to 1998, the population grew at an annual average rate of 2.68% (Balochistan 2.42%).

2 The Balochistan provincial road network comprises 2,351 kilometers (km) of national highways (administered by the National Highway Authority), 8,146 km of provincial highways, 8,659 km of secondary roads, and 3,940 km of access roads.

3 PRs100.9 million was spent on road maintenance in FY2001/02. It is estimated that twice this amount is needed to preserve the road asset base in Balochistan.

4 Karachi and Qasim ports. A new port being constructed in Gwadar in southwest Pakistan is expected to open during 2005.

suffer, however, from lack of or inadequate facilities and systems[5] at border crossings. Moreover, cross border procedures and operations are generally bureaucratic and slow, causing extensive delays in moving goods, and passengers across the borders. The resulting inability to effectively manage and monitor cross-border trade has spurred extensive smuggling and other unauthorized flow of goods, and thus distorts trade and tariff regimes in the subregion. While Pakistan's trade within the Economic Cooperation Organization[6] (ECO) accounts for only 3% of its total trade, Afghanistan is Pakistan's third largest trading partner within ECO after Iran and Turkey. Recent data on trade showed that nearly 90% of Afghanistan's exports of $1.2 billion[7] and about 46% of its imports were through Pakistan. A significant portion of Afghanistan's exports was reexports of imports from Pakistan.[8] Thus, Afghanistan's trade with Pakistan was dominated by unofficial exports and reexports amounting to $941 million, followed by import transit trade ($221 million) and import trade ($61 million). With further progress on securing peace and positive developments in Afghanistan, trade between Afghanistan and Pakistan and transit traffic through Pakistan could substantially increase.[9]

B. Analysis of Key Problems and Opportunities

4. **Poverty.** In Pakistan the incidence of poverty increased from 30.6% in FY1999 to 32.1% in FY2001 leaving about 42 million people in poverty.[10] Poverty is largely a rural phenomenon. About 108 million people live in rural areas (77% of the total population of 130 million). More than 40% of the population live below the poverty line in the rural areas. The overall poverty level in Balochistan is reported at 39.1% (43.7% in rural areas and 29.5 % in urban areas).[11] The Poverty Reduction Strategy Paper of the Government puts the overall poverty figure in Balochistan at 41-47%. About 50% of the rural villages in Balochistan have no or very poor road access. ADB is actively engaged in policy dialogue on poverty reduction and related issues with the federal and provincial governments. The poverty focus is emphasized in the country strategy program (CSP) of May 2002 and reinforced in its update (2004-2006), and in the poverty partnership agreement that was signed in 2002 between the federal Government and ADB. Balochistan provincial government (BPG) is prioritizing improvement of provincial connectivity and enhanced access to reduce poverty. Research and international experience have shown that improved connectivity and road transport play an important role in reducing poverty by providing basic access to education, health, and markets, and improving access to social and economic opportunities, including labor and product markets, schools, and clinics. Improved road connectivity in Balochistan will help the poor overcome geographic isolation, and enable them to participate in community and wider decision making and overcome social, cultural, and physical barriers.[12]

5. **Institutional Capacity and Constraints.** The Communication and Works Department (CWD) in Balochistan, which is responsible for roads and buildings, is structured and functions as a typical public works department. It's character has not changed significantly during recent

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5 Specifically physical infrastructure, equipment, customs, and other administrative systems.
6 Comprises Afghanistan, Azerbaijan, Iran, Kazakhstan, Kyrgyzstan, Tajikistan, Turkey, Turkmenistan, and Uzbekistan.
7 UNDP/World Bank. 2001.*Study: Afghanistan's Trade with Neighboring Countries*. Manila. Refers to official and unofficial trade.
8 This is partly because of the lower prices in Afghanistan and partly because of unofficial and informal trade routes.
9 Currently about 4,000 trucks pass Chamman border crossing every day. Past volume of official transit freight traffic though Pakistan never exceeded 3 million tons per year.
10 This is based on the official poverty line defined at 2,350 calories and analysis in the poverty reduction strategy program (PRSP) of the Government.
11 Talat Anwar and Sarfraz Qureshi. 2001. *Trends in Absolute Poverty in Pakistan*. 1990/91–2000/01. The study is based on a poverty line defined at an adult equivalent caloric intake of 2,550 calories per day.
12 ADB. 2000. *Assessing the Impact of Transport and Energy Infrastructure on Poverty Reduction*. Manila; ADB. 2002. *Impact of Rural Roads on Poverty Reduction: A Case Study-Based Analysis*. Manila.

years. Planning and institutional capacity at for road maintenance at CWD is weak. In addition, the use of road asset management, maintenance systems and procedures are not institutionalized and integrated in the day-to-day operations at CWD. While a separate road management unit was established in 1999 with support of ADB and the Nordic Development Fund, the unit was essentially discontinued in 2001 due to lack of provincial support. Another important but more difficult issue is alignment of staff resources at CWD. Moreover, as a consequence of the ongoing devolution program,[13] significant challenges are confronting BPG, which sets the basis for the transforming CWD and the 26 district administrations. At the federal level, NHA is undergoing a policy and institutional reform program supported by ADB through a program loan component[14] and the World Bank, which aims to transform NHA into a modern road management agency. This program includes (i) institutional development and capacity building at NHA, (ii) improved road maintenance processes, (iii) support for the road maintenance fund that was established in 2002, (iv) a road safety program, (iv) an axle load control program and (v) improved financial management system. Although some initial delays occurred, NHA made good progress during 2002 and 2003 and met the agreed upon first tranche conditions (June 2002) and the second tranche conditions (July 2003), which had been agreed upon with ADB.[15]

6. **Road Maintenance, Safety, and Vehicle Overloading.** During the last 5 years, spending on road maintenance in Balochistan has fluctuated between PRs47.5 million and PRs100.9 milliont.[16] The recent expansion of the provincial road network and the increase in the road asset base indicate that bigger budget allocations for maintenance are required to sustain the provincial road network. This implies a shift from road network expansion to improvement and maintenance of existing roads. It also suggests allocation of more funds for road maintenance and further involvement of the private sector to enhance capacity and efficiency. With respect to road safety, Pakistan's total fatalities-to-vehicle ratio is about 10 times higher than that in developed countries.[17] The situation is worst in Balochistan because of low awareness, poorly trained drivers, and a large number of narrow roads. In 1999, 467 people were killed in road accidents. Reported road accidents reached about 346, but the actual number is likely to be much higher. Overloading of trucks is also a problem on major roads in Balochistan. While NHA has initiated actions to address this growing problem, further efforts by CWD are required to control the increasing damage to its road network. Appendix 3 gives the road sector analysis.

7. **Transit Arrangements.** Transit trade between Pakistan and Afghanistan is governed by the Afghan Transit Trade Agreement (ATTA) signed in 1965. ATTA covers two main routes passing via Chamman and Torkham. Transportation of Afghan cargo is allowed through railway service only. Road transport across Pakistan for certain humanitarian aid shipments bound for Afghanistan is also allowed as an exception by the National Logistics Center. To protect domestic industry and to discourage smuggling, 24 items not allowed under ATTA have been put on the negative list. Recently, however, eight items on that list were deleted and Pakistan has also allowed Port Quasim as an alternative port of entry for transit. Additional measures may include (i) improving the application of current procedures and working arrangements at the main cross-border points, (ii) allowing private operators to carry containers by road, (iii) introducing the use of

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[13] Under the devolution program, local governments at district level have been created.

[14] ADB. 2001. *Report and Recommendation of the President to the Board of Directors on Proposed Loans to the Islamic Republic of Pakistan Road Development Sector Program* . Manila.

[15] The first tranche of the program loan component ($35 million) was released in June 2002 after NHA had met all conditions in the reform program. The second tranche ($15 million) was expected to be released in October 2003.

[16] Road maintenance spending level in Balochistan: FY1998/99? PRs.77.3 million; FY1999/2000? PRs.47.5 million; FY2000/01? PRs.77.4 million; and FY2001/02? PRs.100.9 million.

[17] Pakistan had 18.69 fatalities per 10,000 vehicles in 1992. This was lower than the number for India (25.27) and Bangladesh (85.58), but higher than that in Southeast Asia (5-15 fatalities per 10,000 vehicles) and far above the countries with the lowest number of fatalities, such as the United Kingdom (1.60) and Japan (1.74). Source: ADB. 1995. *Regional Initiatives in Road Safety*. Manila.

the transit-under-bond system (TIR carnet), (iv) adopting measures to discourage illegal trade and smuggling, and (v) improving trade facilitation by revising ATTA with a view to improve the competitiveness of the transit routes through Pakistan. Due to its geographic location, Pakistan has a great opportunity to increase trade by providing efficient transit and competitive access to ports for landlocked Afghanistan if impediments to trade are relaxed and eventually removed.[18] The proposed Project addresses the prevailing constraints by providing TA to review and improve current transit arrangements under ATTA and by improving transit operations at one major border crossing on a pilot basis.

8. **Road Sector Assistance in Afghanistan and Central Asia.** The road network in Afghanistan is in an extremely poor state following decades of war and civil unrest, and road sector institutions are weak or absent. To address these shortcomings, ADB is actively involved in the road sector in Afghanistan. ADB is supporting TA for capacity building[19] and TA to develop a framework and legislation for cross-border transport, covering Pakistan, Afghanistan, and the countries in Central Asia.[20] ADB is also financing rehabilitation and reconstruction of important roads in Afghanistan: (i) rehabilitation of the Kandahar–Spin Bolak road (105 km),[21] which is the main southeast highway connecting Afghanistan with Pakistan and Balochistan at Chamman border crossing, and (ii) rehabilitation of the Pule Khumri–Mazar Sharif–Sheberghan–Andhoy road section (392 km) in the north of Afghanistan including the road link connecting Afghanistan with Uzbekistan (Naibabad–Hairatan road, 55 km). In Central Asia, ADB is also active in supporting subregional cooperation and removing impediments to cross-border road transport through regional TA[22] and financing road rehabilitation projects in Kyrgyz Republic, Mongolia, Tajikistan, and Uzbekistan.

9. **Sector Coordination and Assistance.** ADB, World Bank and Japan Bank for International Cooperation (JBIC) are the principal providers of assistance to the road sector in Pakistan (Appendix 4). ADB has approved seven loans totaling $747 million for Pakistan's road sector, four of which have been for umbrella projects that include components in each of the country's four provinces.[23] The World Bank has mainly focused on assistance to the national road network, specifically rehabilitation of the main domestic artery (Karachi-Lahore-Peshawar), along with support for policy and institutional reform of NHA at the federal level. Following the adoption of a coordinated policy agenda for road sector reform at the national and provincial levels, ADB has targeted (i) development of provincial road networks, (ii) rehabilitation of national highways of regional significance, and (iii) assistance to improve efficiency of movement of goods and passengers across borders in the region. JBIC has also provided assistance to provincial networks, complementing ADB assistance.

10. **Experience in Subregional Cooperation.** ADB has played a catalytic role and is actively supporting several subregional cooperation initiatives in Asia. These initiatives include support for (i) the Greater Mekong Subregion (GMS), which was initiated in 1992; (ii) Central Asia; and, more recently, (iii) South Asia Social and Economic Cooperation (SASEC). Experience gained in these initiatives show that peace encourages the natural process of economic integration, including

[18] Initiatives in regional trade facilitation include (i) the proposed South Asian Free Trade Area (SAARC), and (ii) the Framework Agreement on ECO Trade Cooperation.
[19] ADB. 2002. *Technical Assistance to Afghanistan for Capacity Building for Reconstruction and Development Project*. Manila.
[20] ADB. 2002. *Technical Assistance to Afghanistan for Development of Framework and Legislation for Cross-Border Transport*. Manila.
[21] ADB. 2002. *Technical Assistance (JFPR) to Afghanistan for Reconstruction of Kandahar–Spin Bolak Road (105 km)*. Manila.
[22] ADB. 2002. *Technical Assistance for Regional Cooperation in Transport Projects in Central Asia*. Manila.
[23] Three loans have been for rural (farm-to-market) roads projects, and one is for a provincial highways project.

cross-border trade, investment and labor mobility. ADB also contributes to economic growth, improved employment opportunities, and reduced poverty by tapping the comparative advantages of the participating countries. ADB's experience in subregional cooperation demonstrates progress and success when collaborating on specific and project-driven activities, such as transport infrastructure and cross-border facilitation including transit, within the framework of existing relationships. The proposed Project builds on this experience by introducing a pilot cross-border development initiative at one major border crossing in Pakistan.

11. **Lessons in Project Implementation.** Implementation of the four provincial umbrella projects financed by ADB (para. 9) suffered from fragmented efforts, and coordination problems, which led to project implementation delays. The provincial governments, including BPG, demonstrated, however, their capability to implement the projects as envisaged at appraisal with the support of design and construction supervision consultants. Adopting a provincial focus, the identified weaknesses in the umbrella projects were addressed in subsequent provincial road projects for Sindh and Punjab. Both projects are in the early stages of implementation with recruitment of consultants and procurement actions, and are progressing through with some initial start-up delays. These delays are partly due to the federal and provincial assembly elections, which took place last year and led to changes in the provincial governments. During project preparation, agreement was reached with CWD and NHA that experienced and dedicated staff would be appointed to key project management positions, specifically for positions as project and deputy project directors. Supported by project preparatory TA consultants, NHA and BPG have advanced in preparing the project proforma (PC-1). NHA and BPG have also allocated adequate resources for preconstruction activities and early procurement actions, including prequalification of civil works contractors. Project implementation will be also be helped by strong support from BPG and the chairman of NHA.

III. THE PROPOSED PROJECT

A. Objectives

12. The objectives of the Project are to reduce poverty and support pro-poor economic growth by (i) enhancing access of rural populations to economic opportunities and social services by improving the province's road network, (ii) improving trade facilitation and efficiency of the priority road transport corridor linking Pakistan, Afghanistan, and Central Asia, which crosses Balochistan; and (iii) providing essential institutional support and capacity building at CWD.

13. The Project will improve the inadequate provincial connectivity and allow better access, specifically for the rural poor, to education and health facilities, markets, districts, and the provincial cities. The Project will also, together with ongoing and planned road rehabilitation and reconstruction in Afghanistan, support subregional cooperation by (i) improving cross-border movements of goods and passengers at the Chamman border crossing between Pakistan and Afghanistan, (ii) improving access to ports located in southern Pakistan, and (iii) reducing transport cost through improved road conditions and shorter travel time.

B. Components and Outputs

14. The Project includes four main components: (i) a provincial institutional development component (PIDC) supporting CWD and districts, (ii) a provincial access component (PAC) comprising rehabilitation and construction of about 1,100 km of provincial roads, (iii) a cross-border development component (CDC), and (iv) a national highway component (NHC) comprising rehabilitation of 247 km of national highways. As with the two most recent provincial road

projects, rehabilitation and construction of the provincial roads will follow a sector approach.[24] To enhance the poverty impact of the Project and ensure that development benefits reach the people living along the project roads, a complementary community support program will be provided through a TA loan (paras. 55-56). In addition, a TA grant will assist the Government to review, develop, and improve trade facilitation measures and prepare a program for increased efficiency of cross-border movements of goods and passengers (para. 57).

15. The PIDC supports the policy actions, which have been agreed upon by BPG and ADB and form part of the Project (Appendix 5). These include (i) reviewing CWD's institutional setup and arrangements, including staffing and competence levels; (ii) strengthening the core road management functions in CWD's organizational structure, specifically for planning and executing maintenance; (iii) introducing HDM-4 at CWD and revamping the road asset management system (RAMS); (iv) strengthening road maintenance processes and introducing performance-based maintenance contracts; (v) reviewing the provincial sources for road maintenance funding, including the provincial budgets and availability of funds for road maintenance, and developing a road user charge mechanism and establishing a road fund for secure and stable funding of maintenance; (vi) increasing funding of maintenance; (vii) improving road safety; and (viii) building capacity in environmental and social assessment and resettlement planning.

16. The CDC focuses on trade facilitation and draws on recent findings from the TA (footnote 20) on subregional cooperation, especially with respect to corridor, port, and transport developments. It also draws on the findings of the assessment of the current cross-border agreements, procedures, and practices for the countries in the region, specifically Pakistan, Afghanistan, Iran, Tajikistan, and Uzbekistan. The CDC complements assistance provided by ADB on customs modernization and World Bank assistance that focused on capacity building in trade facilitation. The NHC supports subregional cooperation through improvement of the most utilized but constrained sections of NH25, which form a component of the Central Asia-Afghanistan-Pakistan corridor, but are in poor condition and suffer from capacity constraints and poor road safety. The NHC supports the continued implementation of the ongoing policy and institutional reform program for NHA, supported by ADB and World Bank (Appendix 6).[25]

1. Provincial Institutional Development Component

17. The PIDC draws on the findings of TA provided by ADB[26] to BPG and policy dialogue between ADB and BPG/CWD. The PIDC focuses on building capacity for institutional development and adoption of modern road management practices? specifically for road maintenance? and road safety. The PIDC supports institutional strengthening and capacity building at CWD and conforms with the devolution program that is currently being undertaken in Pakistan. It includes consulting services, with capacity building, training, and provision of essential equipment for CWD.

18. Consulting services will be required to (i) strengthen institutional capacity at CWD and district levels for road maintenance and road safety functions, including training of CWD staff, district staff, and trainers; (ii) introduce modern maintenance concepts? contracting out, performance-based maintenance? and prepare maintenance manuals; (iii) prepare a number of

[24] A sector loan is deemed appropriate where ADB policy for such a loan is met. For this loan, (i) Balochistan provincial government (BPG) has a provincial road sector investment plan to meet the priority development needs, (ii) BPG through the Communication and Works Department (CWD) has the institutional capacity to implement the Project, and (iii) BPG road sector policies can be improved.
[25] The World Bank is preparing a national highway road project, which focuses on continued reform of NHA and upgrading of national highway 5.
[26] ADB. 2002. *Technical Assistance for preparing Balochistan Road Development Sector Project*. Manila.

complete maintenance contracts, including specifications for bidding by CWD on a pilot basis; (iv) further develop the RAMS at CWD and, as appropriate, expand it to district administrations; (v) help BPG and CWD develop (a) a road user charge and mechanism for heavy truck traffic that damages provincial roads, and (b) a mechanism for stable and secured funding of road maintenance; (vi) provide capacity building with training to enhance the institutional capacity of CWD for road safety, social and environment assessments, and handling of social and environmental problems associated with road sector works; and (vii) guide nongovernment organization (NGO) to conduct poverty monitoring of the Project, specifically covering monitoring of contractors' compliance with applicable labor and other laws, and measures to prevent child abuse and promote equal treatment and facilities for both genders.

19. The PIDC will provide a maintenance system and information technology[27] to enhance CWD's delivery of services to districts and people living in Balochistan. These inputs are important for (i) strengthening planning and maintenance functions, and (ii) further developing the RAMS. The PIDC also provides for equipment for traffic and condition surveys, which are deemed important but are currently not available at CWD.

2. Provincial Access Component

20. The PAC comprises (i) rehabilitation and construction of about 19 provincial roads totaling about 1,100 km, and (ii) a periodic maintenance subcomponent. Following a sector approach, the Project will rehabilitate and construct about 6 provincial highways totaling about 662 km and about 13 rural access roads totaling 468 km, which collectively pass through 20 of Balochistan's 26 districts, many of which are poor and are located in less developed areas. Most of the provincial roads are located in poor and deprived districts. [28]

21. The provincial roads will generally be widened to 3.7 or 7.3 meters, depending on traffic density, and provided with a 2.0- or 2.5-meter-wide shoulders on each side. Wherever possible, widening will be confined to one side to minimize disruption or adverse impacts and ease construction and traffic management. The provincial roads will be designed for surface treatment. As existing alignments will be followed, resettlement and environmental impacts will be minimal.

22. The provincial roads were identified and assessed under previous ADB TA financed by loans, and form part of the provincial road master plan. Six (three provincial highways and three rural access roads) of the 53 prescreened roads, totaling about 400 km and representing a sample of typical roads from Balochistan and poor districts, were selected as core roads for further evaluation under the same TA (footnote 26) and subsequent detailed design. The selected core provincial roads were subjected to multiple economic and social criteria analyses: economic analysis, social analysis, assessment of poverty reduction impacts, and environmental examination. A number of stakeholders, specifically from the district administrations, poor people, road users living along or near the roads, and NGOs were involved in the selection process. Interviews and surveys of several stakeholder groups indicate strong support for rehabilitating the selected provincial roads.

23. **Periodic Maintenance Subcomponent.** As a pilot initiative, this subcomponent is intended to finance a 3-year time slice of priority road maintenance activities. For each of the 3 years, this component will cover periodic maintenance of an estimated 500 km gravel and surfaced roads to be carried out through maintenance contracts.[29] The works will be restricted to

[27] Hardware and software.
[28] Based on a district development ranking carried out by the Punjab Economic Research Institute.
[29] The estimated total cost for this subcomponent is $500,000. See Appendix 7.

periodic maintenance works and will comprise mainly grading and leveling of gravel roads. The roads for this component will be selected after completion of prescreening and condition surveys by the consultants under the PIDC. The works will be supervised by the consultants under the PAC. Since this component is included to encourage BPG to commit more of its annual budget to periodic maintenance resources, the cost to be met under the ADB loan will be limited to 30%, while the balance of 70% will be met through BPG counterpart funding. This component will increase awareness and understanding of road maintenance, contribute to securing BPG resources for periodic road maintenance, and address the critical backlog in road maintenance.

24. Under the PIDC, a mechanism for improved axle load control will be designed. In addition, CWD will pursue a dedicated information campaign and training in close coordination with the other BPG departments and agencies responsible for enforcing axle-load control and information, such as the police, drivers' training centers, truck and bus owners and operators, road user associations, and the general public.

3. Cross-Border Development Component

25. The CDC comprises physical and nonphysical subcomponents. The physical component addresses the identified constraints, which specifically include lack of basic infrastructure, and scattered facilities along with inefficient and slow handling procedures at the border crossings. The non-physical component includes a detailed review (under TA accompanying this loan) including training by consultants on current cross-border practices covering immigration, customs, security, and other cross-border functions and related procedures at the four main border points in Pakistan. The review will be the basis for a detailed analysis and preparation of an action plan aimed at simplifying, and relaxing or removing nonphysical and nontariff barriers within the existing legal and regulatory framework. In addition, the review will provide inputs for developing and designing a new cross-border facility at Chamman. The objective is to improve efficiency, reduce transaction cost, improve service, and reduce waiting times at the border crossing. Using the findings and recommendations of the TA, the design and supervision consultants under the NHC will design a new cross-border facility at Chamman, which will be constructed under the civil works contract in the NHC (para. 26).

26. It is envisaged that the new cross-border facility at Chamman will perform most or all main functions required for efficient cross-border movement of goods, vehicles, and passengers. Staff responsible for immigration, customs, and security, and staff providing banking, insurance, and other services will be located in the new facility. The new cross-border facility will be constructed and equipped with modern systems and equipment. It will comprise a new building, truck parking facilities, cargo inspection sheds, weigh bridges, telephones and customs systems. The private sector will be invited to provide services and establish complementary storage and other facilities. It is envisaged that the new concept may, at a later stage, be broadened to a joint project with the Government of Afghanistan with a view that the new facilities might take the form of a one-stop shop operated jointly by the authorities on both sides of the border. The component includes training the various staff; handling immigration, security, customs, weigh stations; and other functions at the cross-border facility, focusing on modern and efficient practices. Immigration and customs officers will also receive training in regulations governing the movement of vehicles, persons, and freight through Pakistan and in and out of Afghanistan, the duties payable on various types of cargo, the nature of any prohibited cargoes, and measures to improve efficiency and reduce time taken to cross the border.

4. National Highway Component

27. For the NHC, which will facilitate regional traffic and also improve connectivity to Balochistan's provincial road network, 247 km of NH25 will be rehabilitated and widened covering the sections (i) Kalat–Quetta, 130 km, and (ii) Quetta–Chamman, 117 km.

28. The national highways will be 7.3 meters wide with a 3.0-meter-wide shoulder on each side. Wherever possible, the widening will be confined to one side to minimize disruption or adverse impacts and ease construction and traffic management. The roads will be designed for asphalt-concrete pavements. As existing alignments will be followed, resettlement and environmental impacts will be minimal. Road safety audit will be conducted as part of the update of the existing designs. A summary of the design approach and technical standards is in Supplementary Appendix A.

29. NHA identified the national highways, which were assessed under previous TA and form part of NHA's medium-term budget framework and approved investment plan. Their feasibilities were further evaluated and confirmed under TA (footnote 26),[30] and subjected to detailed technical evaluation and assessment, economic viability analysis, and analysis of poverty, social, and environmental impacts.

30. The Project will install weigh stations at the Chamman border crossing, along with equipment and related civil works. The location and technical specifications will be determined during the update of the detailed design during implementation of the Project from 2004. As a separate ongoing initiative, NHA is implementing a road safety action plan and an axle-load control program, which also comprises a dedicated information campaign and training of agencies responsible for enforcement and information, including the police, institutions for education, drivers' training centers, truck and bus owners and operators, road user associations, and the general public.

C. Special Features

1. Road Maintenance

31. The Project supports improved road maintenance with several subcomponents and activities under the PIDC. The medium-term road maintenance needs and funding sources will be assessed and a funding mechanism designed and implemented under the Project. This may include design of a road user charge mechanism that will recover damage caused by heavy truck traffic by charging a maintenance fee or something similar for all heavy vehicles passing strategic points on Balochistan's road network. Another initiative supported by the Project is strengthening of maintenance planning capacity building, and improved execution of maintenance by contracting it out to the private sector. The PIDC includes preparation of complete contract documentation, including specifications, for bidding out periodic and routine maintenance to the private sector. In addition, contract documentation with specifications will be developed for performance-based maintenance on a pilot basis. The objective is to increase contracted-out maintenance over time so that the share of contracted-out maintenance to the private sector reaches at least 10% by 2005, 20% by 2006, and 40% by 2007.

[30] The two selected sections were found to be the most constrained and narrow sections on NH25, but carry the most traffic of all sections on NH25. Most other sections on NH25 have also previously been widened to 7.3-meter carriageway plus shoulders.

2. Road Safety

32. Poor road safety affects the many poor people who live in the project influence area. Acknowledging the importance of improving road safety while taking into account the provincial institutional capacity, BPG will prepare, with the support of consultants, a provincial road safety action plan under the PIDC. Representatives of road user groups, NGOs, and others? including the most vulnerable to such accidents, the poor pedestrians? will be consulted. The action plan, which will build on and complement other federal and provincial road safety programs, will comprise a series of preventive measures, such as education, information campaigns and training; and corrective measures, including enforcing traffic safety regulation and introducing road safety audits for provincial roads, to reduce the number of road accidents in Balochistan. The Project will provide support with input from consultants under the PIDC and provision for equipment, such as for accident reporting or other requirements, as might be identified in the road safety action plan. It is envisaged that a reputable NGO will be engaged to administer and coordinate the work, which will be carried out under the action plan, and also to assist in its implementation when feasible and justified. At the federal level, NHA is undergoing a road safety program supported by ADB and World Bank (footnote 14).

3. Capacity Building for Environment and Social Assessment

33. CWD lacks staff, expertise, and capacity to carry out environment and social assessment. To address this constraint, CWD has agreed to establish the Road Environment and Social Assessment Cell (RESAC). RESAC is expected to carry out many functions for the environment, and social safeguards and assessment for the road sector. However, RESAC lacks staff, expertise, and capacity to carry out environment and social assessment. To ensure that RESAC achieves its intended role and function, a program with action to enhance the capacity of this unit will be developed and implemented under the Project. A consultant team on social development, gender, resettlement, and environment under the PIDC will help CWD prepare and implement the institutional enhancement program.

4. Subregional Cooperation

34. The Project is the first in Pakistan, which, together with other ongoing and planned road projects in Afghanistan, supports subregional cooperation through physical and nonphysical improvements of critical sections of NH25, which form part of the transport corridor of Central Asia-Afghanistan-Pakistan that passes through Balochistan. Apart from providing improved access to ports in southern Pakistan, the Project will also facilitate the economic and social development of the transport corridor area through improved access and easier movement of people and goods, and these help reduce poverty and support pro-poor economic growth. Other nontransport-related initiatives, including those by other aid agencies, will provide linkages to realize the corridor concept and support area development along that corridor.

D. Cost Estimates

35. The estimated cost of the Project is $267.3 million equivalent, including civil works, equipment, consulting services, land acquisition, resettlement, institutional support to CWD and the project management unit, taxes, customs, and contingencies. The cost comprises $162.5 million in foreign exchange (60.8% of the total) including $8.0 million for interest during construction, and $104.8 million equivalent (39.2%) in local currency. Table 1 summarizes the cost estimates (detailed cost estimates are in Appendix 7).

Table 1: Cost Estimates
($ million)

Item	Foreign Exchange	Local Currency	Total Cost
A. Base Cost[a]			
1. Provincial Institutional Development Component			
a. Equipment	1.5	0.5	2.0
b. Consulting Services	1.6	0.5	2.1
2. Provincial Access Component			
a. Land Acquisition and Resettlement[b]	0.0	3.0	3.0
b. Provincial Roads Improvement	71.1	46.8	117.9
c. Consulting Services–Detailed Design and Construction Supervision	4.7	2.7	7.4
3. Cross-Border Development Component			
a. Civil Works	1.8	0.7	2.5
b. Equipment	1.0	0.5	1.5
4. National Highway Component			
a. Land Acquisition and Resettlement[b]	0.0	1.0	1.0
b. National Highways Improvement	54.7	36.5	91.2
c. Consulting Services – Construction Supervision	1.5	1.9	3.4
5. Incremental Costs	0.0	4.0	4.0
Subtotal (A)	**137.9**	**98.1**	**236.0**
B. Contingencies			
1. Physical[c]	8.3	3.5	11.8
2. Price[d]	7.4	3.2	10.6
Subtotal (B)	**15.7**	**6.7**	**22.4**
C. Front-End Fee	**0.9**	**0.0**	**0.9**
D. Interest During Construction[e]	**8.0**	**0.0**	**8.0**
Total[f]	**162.5**	**104.8**	**267.3**

[a] In mid-2003 prices.
[b] Including fund for noncore subproject roads.
[c] 10% of base cost, excluding provincial roads improvement component.
[d] At 2.4% annually, excluding provincial roads improvement component.
[e] Including commitment charge for ordinary capital resources loan.
[f] Including taxes, customs, and duties.
Source: Consultants and Asian Development Bank staff estimates.

E. Financing Plan

36. The Government has requested a loan of ¥20,266,370,000 ($185.7 million equivalent) from ADB's ordinary capital resources to help finance the Project (Table 2). The loan will have a 25-year term including a grace period of 5 years, an interest rate determined in accordance with ADB's London interbank offered rate (LIBOR)-based lending facility, a commitment charge of 0.75% per annum, a front-end fee of 0.5% (the fee will be capitalized in the loan), and such other terms and conditions set forth in the draft Loan and Project Agreements. The Government has provided ADB with (i) the reasons for its decision to borrow under ADB's LIBOR-based lending facility on the basis of these terms and conditions, and (ii) an undertaking that these choices were its own independent decision and not made in reliance on any communication or advice from

ADB. The Borrower will make available the loan proceeds to NHA ($85.7 million equivalent) and BPG ($100 million equivalent).

Table 2: Financing Plan
($ million)

Source	Foreign Exchange	Local Currency	Total Cost	Percent
ADB	162.5	23.2	185.7	69.5
Government	0.0	81.6	81.6	30.5
Total	**162.5**	**104.8**	**267.3**	100.0

ADB = Asian Development Bank.
Source: ADB staff estimates.

37. The amount of local currency funding takes account of the current fiscal situation in Pakistan. The country is one of the poorest in the world and the federal Government and BPG have difficulty mobilizing sufficient funds from their own resources to meet all development commitments, including rehabilitation and upgrading critical physical and social infrastructure. Although the need to generate domestic resources to finance investments for sustaining economic growth is well recognized, the country has not yet developed the resource base necessary to adequately finance without external assistance the required level of investments. Under the current circumstances, a large investment-savings gap will continue exerting pressure on development program. Thus, external support is needed to meet some of the local currency costs of projects. In line with ADB's policy to help provide local currency cost financing, particularly for projects that address priority poverty concerns in economically less developed regions, about 12.5 % of the loan will be for that purpose.

F. Implementation Arrangements

1. Project Management

38. CWD will be the Executing Agency for the PIDC and the PAC (Supplementary Appendix B). The secretary of CWD will assume overall responsibility for project implementation. The additional chief secretary for development will chair a steering group comprising senior staff of BPG and CWD, which will guide the CWD institutional strengthening program and the action plan forming part of the policy matrix agreed upon with ADB. The core project management unit, (PMU) through the Project Engineering Cell, will be responsible for project management. The PMU already established by CWD is headed by a project director. The project director will be supported by other staff including two deputies who will be in charge of day-to-day administration of the provincial roads. While the PMU has experience with ADB-financed projects and is generally capable, some institutional strengthening is required to ensure effective implementation. Thus, the PMU will be supported by institutional consultants under the Project as well as by design and construction supervision consultants. The institutional consultants will support the PMU in project management, supervision of short-term consultants under the PIDC, contract administration, and project preparatory work, while the design and supervision consultants will be responsible for construction supervision in accordance with the Federation International des Ingenieurs-Conseils (FIDIC) conditions. The consultants will help build capacity at CWD and PMU to effectively manage and implement all project components, specifically the civil works and resettlement plans, and prepare subprojects for the provincial roads component. Early in their work, the institutional consultants will prepare project preparation and implementation manuals. The implementation arrangements for the PIDC and the PAC are shown in Appendix 8.

39. NHA will be the Executing Agency for the CDC and the NHC (Supplementary Appendix C). The NHA Chairman will chair a steering group, comprising senior staff members of NHA, who will guide in implementing the NHC. While NHA will be responsible for design and construction of the new cross-border facility under the CDC, the secretary of the Ministry of Communications (MOC), in consultation with the existing Task Force on Trade Facilitation (chaired by Joint Secretary, Ministry of Commerce), will be the Implementing Agency for the other parts of the CDC, specifically for planning the structure, authority and procedures for operating the new cross-border facility. NHA has already appointed a project director and has experience with ADB-financed projects. It is capable of implementing the Project with support of the construction supervision consultants. The implementation arrangements for the CDC and the NHA are shown in Appendix 9.

2. Subproject Preparation for the Provincial Access Component

40. The core provincial roads are prescreened and meet criteria that are acceptable to ADB. Specifically, the core roads must have been found technically feasible, economically justified, and socially and environmentally responsible. Most of the data needed for selecting and prioritizing these roads were prepared under project preparatory and technical assistance and loan-financed TA. The full data and documentation required for approval of all subprojects are not yet available and data will be prepared under the Project. While the subproject roads will be selected at the discretion of BPG, ADB approval will be required before tendering for road improvement.

41. CWD will submit to ADB only subprojects that meet the criteria for financing under the loan (Appendix 10). The list of prescreened candidate subprojects is in Supplementary Appendix D. Each subproject should (i) be economically viable, (ii) be technically feasible, (iii) be environmentally and socially responsible, (iv) be screened for environmental impacts, (v) be selected based on prioritized district development ranking, (vi) be selected and designed to avoid or minimize resettlement (a resettlement plan should be prepared where needed), (vii) be part of the provincial road master plan, (viii) be in compliance with all applicable ADB guidelines, (ix) have received all BPG and other approvals, (x) have sufficient counterpart funding, (xi) have budget provided for operation and maintenance of the road, and (xii) have district commitment to maintenance of that road. CWD will seek ADB approval by submitting a request for subproject approval that will include (i) a cover sheet; (ii) a basic data sheet; (iii) a location map; (iv) an initial environmental examination (IEE) or environmental impact assessment, as appropriate, in ADB format; (v) the BPG environment approval; and (vi) a resettlement plan, in a format acceptable to ADB.

3. Implementation Schedule

42. The Project will be implemented over 6 years, including preconstruction activities. A summary implementation schedule, based on a detailed schedule prepared by NHA, CWD, and ADB, is in Appendix 11. The schedule assumes that the project loan will be approved in November 2003 and be effective by 1 February 2004. The two national highways and the six core provincial roads will be ready for contract awards in June 2004. The subprojects for the noncore provincial roads will be implemented in subsequent phases, going through subproject preparation, bidding and award of civil works, and construction.

4. Procurement

43. Goods and related services and civil works will be procured in accordance with ADB's *Guidelines for Procurement.* Civil works for the national highway component will be procured through four contract packages following international competitive bidding (ICB) among

prequalified bidders. ICB will be followed for all civil works contracts with an estimated value exceeding $3 million. Civil works contracts with an estimated value equal to or below $3 million will follow local competitive bidding procedures acceptable to ADB, with separate prequalification. Contracts for goods and related services estimated to cost in excess of $500,000 will be subject to ICB. The provincial roads entail about 12 contracts for civil works, each with an estimated value of $1.0 million to $10 million. Equipment will be procured under international shopping procedures. International shopping procedures will be followed for supply contracts valued at less than $500,000 equivalent and direct purchase procedures will be followed for items valued at less than $100,000. NHA and CWD have agreed to include the relevant sections of ADB's *Anticorruption Policy*[31] in all bidding documents and contracts. In July 2003, ADB approved CWD's request for advance action for procurement. BPG and CWD were advised that such approval does not commit ADB to finance the Project. Contract packages for the national highways and the core provincial roads are in Appendix 12.

5. Consulting Services

44. The Project will require 250 person-months of international and 4,509 person-months of domestic consulting services for (i) institutional development with capacity building under the PIDC, (ii) construction supervision of the civil works for the national highways, and (iii) detailed design and construction supervision for civil works for the provincial roads. All consultants will be selected in accordance with ADB's *Guidelines on the Use of Consultants,* and other arrangements satisfactory to ADB for engaging domestic consultants. The quality- and cost-based selection procedure will be followed in selecting consulting firms. Both NHA and CWD have agreed to include the relevant sections of ADB's *Anticorruption Policy* in all invitations for bidding documents and contracts.

45. ADB and CWD have agreed on the objectives and the scope of the consulting services for institutional development and capacity building under the PIDC. The scope includes support and capacity building in project management, road maintenance and development of a RAMS, and support services to the PMU including preparation of subprojects in a format acceptable to ADB. Consulting services will be provided by an international firm in association with a domestic firm. The approximate staffing inputs will be about 78 person-months of international and 200 person-months of domestic consultants. The overall period of the services will be about 48 months. The summary terms of reference are in Supplementary Appendix E.

46. CWD will engage one consulting firm for design and construction supervision of provincial roads. Staffing inputs will be about 101 person-months of international and 3,202 person-months of domestic consultants. The overall period of services will be about 72 months. Furnished office facilities, utilities, vehicles, and equipment required by the construction supervision consultants (CSCs) will be provided under the Project. Field offices, material testing laboratories with utilities, and vehicles required by construction inspectors will be supplied through the civil works contracts for the provincial roads, as appropriate. The summary terms of reference are given in Supplementary Appendix F.

47. NHA will engage one international firm in association with a domestic firm for construction supervision of national highways. Staffing inputs will be about 71 person-months of international and 1,107 person-months of domestic consultants. The overall period of the services will be about 72 months. Furnished office facilities, utilities, vehicles, and equipment required by the CSCs will be provided under the Project. Field offices, material testing laboratories with utilities, and vehicles required by construction inspectors will be supplied through the civil works contracts

[31] ADB. 1998. *Anticorruption Policy.* Manila. Available: http://www.adb.org/work/Policies/Anticorruption.

for the national highways, as appropriate. The summary terms of reference are in Supplementary Appendix G.

6. Advance Action

48. On 14 July 2003, ADB approved advance action for recruiting project consultants and for preconstruction activities for the civil works. The Mission has advised NHA and BPG that any such approval would not commit ADB to finance the Project.

7. Disbursement Arrangements

49. All disbursement procedures under the loan will follow ADB's *Loan Disbursement Handbook* (January 2001). Direct payment procedures will be followed for consulting services and the civil works under ICB. Imprest account procedures will be used for the local bidding for civil works for CWD and also for CWD's incremental expenditures under the loan. The imprest account will be established in the National Bank of Pakistan. The initial advance will not exceed the estimated expenditures for 6 months or 10% of the loan amount, whichever is lower. The statement of expenditure procedure will be used for all payments and contracts under $50,000 equivalent.

8. Accounts, Audit, and Reports

a. Accounts and Audit

50. NHA and CWD will maintain records and accounts adequate to identify the goods and services financed by the loan proceeds, the financing sources received, the expenditures incurred, and use of local funds. The accounts will be set up in accordance with sound accounting principles. External accounting experts will be engaged to design project accounts to be used by NHA and CWD. Consolidated project accounts and related financial statements will be audited annually by private sector auditors acceptable to ADB. The audited reports and related financial statements in English will be submitted to ADB not later than 6 months after the end of the fiscal year to which they relate. The director-general of audits will coordinate all account activities and ensure compliance with ADB's audit and accounting requirements. ADB will review and follow up on these matters as necessary.

51. All procurement and disbursement activities will be subject to independent performance audit to ensure transparency, and objective and independent assessment of such activities. The performance audit will be conducted three times during project implementation in conjunction with the annual audit of project accounts and financial statements and will be carried out by the same external auditors. In addition, ADB will conduct project procurement and disbursement audits during implementation as part of its regular review. Both NHA and BPG have agreed to combat corruption in the road sector, including the procurement of contracts, by preparing an anticorruption strategy for the road sector, in consultation with civil society and anticorruption agencies in Pakistan and ADB. The strategy will be prepared within 6 months of loan effectiveness. NHA and BPG will implement the strategy during the remaining project implementation period.

b. Reports

52. Before starting the civil works under the Project, NHA and CWD will provide ADB with monthly progress reports on the status of preconstruction activities. These reports, due at the end of each month, will be sent in electronic form. When the PMUs and the CSCs have been appointed,

they will prepare monthly progress and quarterly reports[32] for NHA, CWD, PMUs and ADB. The format of the reports will be discussed and agreed upon before consulting services starts. Once the works have been completed, NHA and CWD will arrange for the consultants to submit a project completion report within 3 months of the physical completion of the Project.

9. Project Performance Monitoring and Evaluation

53. Both NHA and CWD, with the assistance of the PMUs and the CSCs, will separately carry out project performance and benefit monitoring and evaluation (M&E) by compiling and analyzing appropriate traffic, socioeconomic, and other data for the project roads. This evaluation will follow ADB's *Project Performance Management Systems Handbook.* For this purpose, a baseline survey for one national highway and three provincial roads (comprising provincial highway, secondary and/or access road) will be undertaken. This will be followed by data collection and other surveys during implementation and immediately after these project roads have been completed and handed over. The findings and supporting data will be incorporated into the project completion report. Systems for recording data and statistics for such monitoring will be established as part of the project implementation arrangements with assistance of the PMUs and the CSCs. NHA and CWD, assisted by the consultants will evaluate the benefits of the Project in accordance with a schedule and terms of reference to be agreed upon with ADB.

10. Project Review

54. ADB will conduct regular reviews throughout project implementation. Midway through Project implementation (end-2006), ADB, NHA, and CWD will review the Project and its implementation status in detail. The review will cover the entire scope of the Project to determine whether any adjustments in the proposed procurement and management arrangements are needed. The review will also address any outstanding procurement problems, and financing and scheduling matters.

IV. THE TECHNICAL ASSISTANCE LOAN

55. To enhance the poverty impact of the Project, and to ensure that maximum possible development benefits reach the people living along the project roads and that the resources they rely on are protected and developed in a sustainable way, additional provincial support will be provided through a TA loan. The TA will support devolution and assist BPG and the district administration in enhancing the project benefits to the poor. The TA will have three parts: (i) developing community infrastructure and empowering communities, and (ii) strengthening of social protection and education, and (iii) monitoring and measuring poverty and social impacts. The initiatives under the TA loan are to be selected in a participatory process involving the communities and the district administration from the provincial roads, which are located in the poorest districts in Balochistan. The TA loan will be implemented over 4 years, with expenditure commencing at the start of FY2004. The objectives, tentative scope, financing, and implementation arrangements are in Appendix 13. The TA loan will be executed by CWD and implemented by reputable NGOs, with CWD support and close coordination with the district administrations.

56. The total cost of the TA loan is estimated to be about $1.25 million. ADB will finance all foreign exchange cost and a portion of the local currency costs. ADB will provide one loan equivalent to SDR 701,000 million [$1.0 million equivalent] from its Special Funds resources with a maturity of 32 years, including a grace period of 8 years, and with an interest rate of 1.0% per annum during the grace period and 1.5% thereafter; and such other terms and conditions as are

[32] The quarterly report will also include data on project performance monitoring and benefit monitoring.

substantially in accordance with those set forth in the draft TA Loan Agreement. Consulting services may be provided by one or more consulting firms or individual consultants recruited in accordance with ADB's *Guidelines on the Use of Consultants* and other arrangements satisfactory to ADB.

V. THE TECHNICAL ASSISTANCE GRANT

57. To support the Project and assist in improving trade facilitation, the Government has requested TA to prepare a program for more efficient cross-border operations and to establish a pilot cross-border facility. The total cost of the TA is estimated at $630,000, comprising foreign exchange of $285,000 and local currency of $345,000 equivalent. The Government has requested ADB to finance $500,000 equivalent, including the entire foreign exchange cost and $215,000 equivalent of the local currency cost; the Government will finance the balance of the local currency cost. The TA will be financed on a grant basis by ADB's TA funding program. The TA scope, implementation arrangements, outline terms of reference, cost estimates and financing plan are outlined in Appendix 14.

VI. PROJECT BENEFITS, IMPACTS, AND RISKS

A. Economic Benefits

58. The economic analysis (Appendix 15) of the national highways shows economic internal rates of return (EIRRs) of 37.7% and 40.4%, respectively. The economic analysis of the six core provincial roads shows EIRRs of 14.8–48.9%. The sensitivity analysis shows that the EIRRs for all evaluated roads are above 12% and quite robust.

59. Economic analysis was carried out using the highway design and maintenance (HDM) software. The model evaluates project investments considering capital and recurrent costs as well as the benefits derived from savings in road user costs for the with-project and without-project cases. Economic analysis assumed the project life to be 20 years. Two improvement options were considered, i.e., asphaltic concrete overlay and surface dressing for different permutations of traffic levels and road roughness conditions. The main quantifiable benefits of the project are (i) savings on vehicle operating costs (VOC) for existing traffic, and (ii) VOC savings for generated traffic because of improvements in road surface conditions. In addition, nonquantifiable benefits will arise from improved access to markets, which will boost agricultural economic activities; reduced agricultural input costs because of better transport facilities; improved access to health, education, and other social sector facilities; and subregional benefits arising from improved trade facilitation and reduced delays at border crossings.

B. Poverty Reduction Impact

60. The overall poverty level in Balochistan is reported at 39.1% of which 43.7% is for rural areas and 29.5% for urban areas (para. 4). While the Poverty Reduction Strategy Paper of the Government reports the overall poverty figure in Balochistan at 41-47%, about 50% of the rural villages in Balochistan have no or very poor road access. Around 76% of the population in Balochistan live in rural areas, where the incidence of poverty approached almost 50% before the recent drought, and now may be approaching 70% in drought stricken rural areas, compared with about 35% of the rural population living below the poverty line for Pakistan as a whole. Data for Balochistan is questionable as the low population density makes sampling difficult, but the household survey results indicate that poverty levels among households sampled for the provincial roads are on average 53%.

61. The Project is classified as a poverty intervention and will directly reduce poverty by lowering the costs of transport for the poor traveling to farmers' markets, labor markets, health and education centers, and other social and public services. According to the conducted poverty assessment, the Project will have a large effect on reducing poverty, directly benefiting approximately 500,000 people in 19 districts living adjacent to the roads. Of these, an estimated 200,000 poor people will be direct beneficiaries. Many more will benefit in the long run from the lower transport costs and induced economic and social development.

62. The estimated poverty impact ratios of the core roads range from 26% to over 100% of the net economic benefits accruing to the poor for the Project as a whole, with most roads having between 30% and 50%. Such benefits will become evident in higher agricultural product profit margins for poor farmers, as well as lower cost and more timely access to schools, medical clinics, and employment centers. Furthermore, the lower transport costs will increase mobility among the poor and help them find employment opportunities in other areas.

63. The Project will generate significant employment opportunities for skilled and unskilled labor during implementation. Unskilled labor (males and females) will be employed directly in road construction and maintenance and indirectly by providing materials and services to the construction and maintenance activities. The Project will generate for unskilled labor employment opportunities of approximately 22,560 person-months for national highways and 31,560 person-months for provincial and rural roads. The Project will have positive externalities, which will improve the livelihood of the poor. The poverty reduction aspects of the Project will be enhanced by encouraging contractors to engage local workers in project areas and providing first priority to the poor and disadvantaged in the actual construction of provincial roads.

C. Social Dimensions and Impacts

64. The social dimensions of the Project, including stakeholder participation and equity consideration, are designed to facilitate more pro-poor and equitable distribution benefits. The Project will adopt a participatory approach to effectively implement the resettlement plan and an awareness campaign on road safety issues. The initial poverty and social assessment did not identify indigenous people or minorities because of the insular nature of some communities, and an indigenous people's development plan (IPDP) is not required. Small groups of affected people were interviewed to determine their aspirations and their likely reaction to the road. Support for the roads was universal, with all groups identifying positive outcomes. The summary poverty reduction and social strategy is in Appendix 16.

65. As part of institutional reform, the Project will strengthen the capacity of CWD to address poverty, environment, social safeguard, and social development issues in the transport sector. Gender awareness training will be given to CWD staff at provincial and district levels. Gender-disaggregated data will be collected through the baseline project performance management system survey as well as in subsequent surveys to be conducted at midterm and completion of the project.

66. In compliance with Government and international core labor standards, CWD will be strengthened to monitor the implementation of labor laws, particularly the issues of equal pay for equal work between women and men workers; prohibition of child labor; and provisions on health, sanitation, and appropriate working conditions, including accommodation, where appropriate, for construction workers at campsites during the construction period. A number of covenants related to this issue will be incorporated in the loan agreement.

67. The Project was developed with a view to avoid or minimize the need for land acquisition and involuntary resettlement. A full census was conducted of all affected people, land, structures/assets on the right-of-way for the two national highways and the six core representative subprojects for the provincial roads component, including a 25% socioeconomic sample survey for resettlement planning. The Project will require acquisition of approximately 42 hectares for the two national highways and six core provincial roads, and will likely affect about 487 people as a consequence of land acquisition. Widening and improvement will affect 143 structures (80 shops or small business enterprises, 39 residential houses, and 24 other structures). The estimated cost for land acquisition, compensation, implementation, and monitoring is approximately $4 million equivalent. A summary resettlement plan is presented in Appendix 17. The summary in local language and in English will be distributed to the affected people by NHA and BPG and through NHA and ADB web sites. The complete plan will be publicly available at NHA, CWD, and at ADB's Pakistan Resident Mission. The remaining subprojects (provincial roads) will be selected during implementation. The requirements of ADB's policy on involuntary resettlement for preparing the sector loan necessitated a resettlement framework (Appendix 18) for all subprojects. A resettlement plan for each remaining subproject will be prepared and approved by BPG and submitted to ADB for approval.

D. Environmental Impacts and Measures

68. In accordance with ADB's requirements on environmental assessment, an IEE was prepared for the two national highways and the six core provincial roads. The IEE (Appendix 19) shows that, generally, the Project will not cause significant environmental problems and with the mitigation measures, potential adverse impacts are manageable. The Project will comply with all ADB environmental assessment requirements. The environmental mitigation measures during the construction phase will be incorporated into the bidding documents for civil works, and NHA and CWD will be responsible for overall environmental management in their respective components.

E. Project Risks

69. A potential risk is the delays in project preparation and start-up activities and necessary approvals by NHA and BPG. This risk will be mitigated through approval of advance actions by NHA and BPG and adequate allocation of staff and other resources. The PMUs are already established at NHA and CWD and have been sufficiently staffed and empowered with experienced staff. An implementation unit is also being established at MOC for execution of the TA grant.

70. Another risk is lack of project sustainability due to inadequate maintenance. This risk has been addressed by the asset management and maintenance activities forming part of the Project. NHA and BPG are committed to provide sufficient and increased funding for road maintenance, including increasingly contracting out maintenance to the private sector. Maintenance forms a central and integral part of the PIDC and NHA and BPG will take the necessary steps to increase funding for maintenance and introduce new maintenance concepts and procedures.

71. NHA and BPG are committed to reform of their institutions. While NHA is undergoing a reform program supported by ADB, BPG has agreed with ADB on a set of institutional and other reform measures, covering institutional development with capacity building and improved maintenance and road safety, with measurable targets. Consultants will support BPG and CWD to introduce institutional reform measures and to build capacity. The risks of failure or slowdown in reform are therefore considered manageable.

72. The recent history of conflict in the region and related incidents or threats is a potential risk for the project. The federal and provincial governments are aware of this risk and have taken measures to manage it as far as is possible.

VII. ASSURANCES

A. Specific Assurances

73. In addition to the standard assurances, the Government, NHA, BPG, and CWD have given the following assurances, which are incorporated in the legal documents:

74. **Environment**. NHA and BPG will ensure the following:

(i) Road construction and maintenance will be carried out as required under the Government's environmental laws and regulations and ADB's environmental guidelines including (a) appropriate selection of quarry and borrow sites and their rehabilitation after use; (b) proper disposal of soils and construction materials; (c) use of sound environmental design and construction techniques that ensure slope stability and drainage; and (d) minimization of construction impacts such as dust, diversion of stream flow and increased turbulence, and equipment noise.

(ii) When project roads pass through or near protected or sensitive areas (e.g., nature reserves, parks, conservation areas, biodiversity reserves, or cultural and historical sites), rapid assessment will be carried out before any construction work starts. BPG will ensure cooperation with the responsible authority in preparing and implementing protective measures, which may include area demarcation, fencing, checkpoints, speed bumps, wildlife passage zones, patrols or other measures to protect the area from encroachment or environmental degradation due to road improvement and increased access.

75. **Resettlement**. NHA and BPG will ensure that land acquisition and resettlement activities are implemented in accordance with all applicable laws and regulations, and ADB's policy on involuntary resettlement, as set out in the agreed-upon resettlement plans for core subprojects and resettlement framework (Appendix 18) for noncore subprojects, including the following: (i) land and rights-of-way will be acquired in a proper and timely manner; (ii) compensation will be provided at replacement cost and entitlements as stipulated in the resettlement plan, with ADB's policy to prevail in case of any difference with the Government's laws and regulations; (iii) counterpart funds and disbursements will be provided promptly to affected people, with agreed-upon compensation given to them before any land is taken or civil works contracts awarded, and the Project will not use any emergency provisions that allow land to be taken in advance of payment; (iv) NHA and CWD have guaranteed to meet unforeseen obligations in excess of budget estimates; (v) NHA and CWD will implement adequate supervision, monitoring, and reporting; (vi) a competent independent agency will be in charge of external M&E; (vii) resettlement fund disbursements and expenditures will be audited annually; (viii) adequate information dissemination to and consultation with affected people will be done; (ix) consultation and grievances will be documented; (x) resettlement plans will be updated if the scope of the Project is changed; and (xi) progress will be regularly reported to ADB.

76. **Grievance and Redress Mechanism.** The Borrower will cause CWD and NHA to form within 6 months of loan effectiveness an independent committee consisting of individuals unrelated to the Project to hear any grievance or other complaint relating to resettlement issues.

The committee will hold a public hearing of any grievance or complaint filed with it and make its recommendations for redress, if any, within 45 days of receipt of the complaint.

77. **Health Risks and Provisions.** NHA and BPG will ensure that the civil works contracts include an information and education campaign on sexually transmitted diseases and HIV/AIDS[33] for construction workers as part of the health and safety program at campsites during the construction period. NHA and BPG will ensure that the civil works contracts include legally mandated provisions on health, sanitation, and appropriate working conditions, including accommodation, and clean drinking water for construction workers at campsites during the construction period.

78. **Illegal Trafficking of Women, Children, and Goods.** The Borrower will ensure that NHA and CWD enact measures, satisfactory to ADB, to prevent, monitor, detect, and arrest illegal trafficking of women and children and illegal movement of goods. NHA and CWD will submit to ADB quarterly reports summarizing issues addressed, progress made, and future plans for action.

79. **Labor Laws.** NHA and BPG will ensure that the civil works contractors comply with all applicable labor and worker safety laws, and do not employ children in construction activities. NHA and BPG will set employment targets for women for road construction activities acceptable to ADB and will cause contractors to adhere to such targets. The Government will provide equal opportunity for women for road construction activities, and will not allow contractors to differentiate wages for men and women for work of equal value. A specific clause to this effect will be included in bidding documents, and compliance will be strictly monitored during project implementation.

80. **Anticorruption.** Consistent with its commitment to good governance, accountability, and transparency, ADB reserves the right to investigate directly, or through its agents, any possible financial or management impropriety in conducting the Project. The Government, NHA, BPG, and CWD agree to fully cooperate with any such investigation and extend all necessary assistance, including access to all relevant books and records as well as engagement by NHA or CWD of independent experts that may be needed for satisfactory completion of such investigations. All external costs related to the investigations will be borne by the Project. Within 6 months of loan effectiveness, BPG and NHA will also prepare an anticorruption strategy, which will be implemented during the project implementation period.

81. **Project Management.** In order to ensure continuity, transparency, efficiency and good management, NHA, BPG and CWD will consult with ADB on matters related to management of the Project. All matters relating to selection or appointment, or any changes to such selection or appointment, of any project director or deputy project director, will be duly notified to ADB.

82. **RESAC**. CWD will establish RESAC and provide a core staff including two experienced experts by December 2003 to deal with issues such as resettlement planning and management, poverty reduction monitoring, labor practices, gender analysis, and environment.

83. **Road Sector Reform and Institutional Strengthening.** BPG and NHA give these assurances:

(i) BPG will review CWD's institutional capacity and staffing, and consult with ADB on recommended measures and actions to transform CWD into a modern road

[33] HIV/AIDS = human immunodeficiency virus/acquired immunodeficiency syndrome.

management agency, and other actions as agreed upon in the policy matrix and action plan (Appendix 5).

(ii) NHA will continue the reform of its organization, institutional structure, and other key areas in accordance with its policy and institutional reform program (Appendix 6).

84. **Road Maintenance.** Assurances on road maintenance are made by NHA and BPG.

(i) NHA and BPG will ensure that the improved roads are maintained to design standards and in accordance with sound maintenance practices, that appropriate road maintenance procedures and annual maintenance plans are prepared for the improved national highways and provincial roads, on the basis of modern, agreed-upon maintenance standards, traffic volumes, and assessment of needs.

(ii) Upon completion of the national highways, NHA will contract out periodic and routine maintenance for these roads to the private sector.

(iii) BPG will (a) increase its spending on road maintenance by 5% yearly in real terms (base value: PRs.97.8 million in 2002/03); (b) award three maintenance contracts (covering at least 500 km in total) to the private sector within 24 months of loan effectiveness; (c) award at least one performance-based maintenance contract (covering at least 50 km) to the private sector within 24 months of loan effectiveness; and (d) conduct a study on road user charges and suitable sources and mechanisms for increased funding of road maintenance, and consult with ADB on implementing the recommended measures.

85. **Private Sector Participation.** The Borrower will cause CWD to contract out road maintenance to private sector operators. Such contracts will attain the following percentages of all maintenance contracts awarded annually: 10% by 31 December 2005, 20% by 31 December 2006, and 40% by 31 December 2007. Within 4 months of each of the dates specified, CWD will confirm to ADB the actual percentages attained.

86. **Transit Agreement.** Within 6 months of the receipt of the consultants' report, the Government and ADB will hold a joint workshop to review the report and identify areas for action by the Government, which may facilitate increased trade between Pakistan and Afghanistan through the Chamman border crossing.

87. **Cross-Border Facility.** Within 3 months of the delivery to MOC of the consultant's recommendations and plan for the proposed new cross-border facility, the Government will approve that proposal; within 24 months after the Government approves the plan, NHA will complete construction and installation of all necessary equipment in accordance with the plan; within 36 months of loan effectiveness, the Government will have started operating the cross-border facility at Chamman. Among other things, the cross-border facility will provide a health information service that will include disseminating information to prevent HIV/AIDS and other sexually transmitted diseases.

88. **Subproject Approval Process and Selection.** All subprojects will comply with ADB policies and guidelines, and satisfy ADB procedures for subproject preparation with respect to technical, operational, economic, social, indigenous people, resettlement, and other requirements. Balochistan will make available sufficient counterpart funding for each subproject.

A subproject will be eligible for financing under the loan if it satisfies the criteria set forth in Appendix 10.

B. Condition for Loan Effectiveness

1. The Government, NHA, and BPG will have approved the Planning Commission Proforma I (PC-I) for the Project.

C. Condition for Loan Disbursement

2. No loan proceeds for financing a civil works contract for a national provincial highway or rural road will be disbursed until the CSCs for the national highways or the provincial roads, as appropriate, have been engaged.

VIII. RECOMMENDATION

3. I am satisfied that the proposed loan and the proposed TA loan would comply with the Articles of Agreement of ADB and recommend that the Board approve:

(i) the loan of ¥20,266,370,000 to the Islamic Republic of Pakistan for the Balochistan Road Development Sector Project from ADB's ordinary capital resources with interest to be determined in accordance with ADB's LIBOR-based lending facility; a term of 25 years, including a grace period of 5 years; and such other terms and conditions as are substantially in accordance with those set forth in the draft Loan and Project Agreements presented to the Board; and

(ii) the loan in various currencies equivalent to Special Drawing Rights 701,000 to the Islamic Republic of Pakistan for the Community Development and Poverty Reduction component of the Project, with an interest charge at the rate of 1% per annum during the grace period and 1.5% thereafter; a term of 32 years, including a grace period of 8 years; and such other terms and conditions as are substantially in accordance with those set forth in the draft Technical Assistance Loan and Project Agreements presented to the Board.

Tadao Chino
President

28 October 2003

PROJECT FRAMEWORK

Design Summary	Performance Indicators/Targets	Monitoring Mechanisms	Assumptions and Risks
Goal Enhanced economic growth and reduced poverty in Balochistan and neighboring areas	Increased gross domestic product Reduced poverty	Provincial economic data and statistics	Complementary investments and initiatives of other economic infrastructure, social services and programs.
Purposes 1. Improve connectivity and access to social services and economic opportunities for the poor in Balochistan 2. Increase efficiency, reduce cost and time for movement of goods, vehicles, and passengers between Pakistan and Afghanistan and Central Asia	Improve year-round access to rural areas. Reduce travel times by at least 25% upon completion of project roads Increase cross-border movement of freight and passengers Reduce transport costs for cross-border freight movement by at least 25% by 2009	Progress reports Provincial economic indicators and statistics Project performance monitoring Project completion report Cross-border statistics Asian Development Bank (ADB) review missions	Neighboring governments are strongly committed to cross-border trade. Economic conditions favor increased trade. Other unforeseen barriers to trade do not exist nor are they imposed. Maintenance funding is adequate and sustained. Regulations to prevent overloading of trucks are adequately implemented. Stability, regional peace, confidence building, and development of relationships continue. The Government commitment to regional economic cooperation and trade is strong.
Outputs 1. Provincial institutional development a. Institutional strengthening and organizational changes at Communication and Works Department (CWD) b. Road user charges introduced, road management practices and redesign processes improved c. Strengthened capacity in planning and road maintenance at CWD and at districts d. Strengthened axle-load control e. Improved road safety f. Increased private sector involvement g. Increased involvement of road user associations in various areas of provincial road development 2. Improved 1,100 km of provincial roads in Balochistan 3. New cross-border facility at Chamman constructed and operational	As per policy matrix and action plan agreed upon with ADB Project review missions By 2006 Accident reduced by 50% in 2008 Increased 40% by 2007 (Base year 2003) Completed by June 2008 at appropriate quality and cost Completed in June 2006	Monthly progress reports ADB review missions Audited annual financial statements Project completion reports Benefit monitoring and evaluation	Involved agencies and stakeholders are in close coordination. Timely actions are implemented as agreed upon with ADB. Maintenance practices are weak. Reform program is implemented as scheduled. Award of civil works contracts is timely. Award of civil works contracts is timely.

Design Summary	Performance Indicators/Targets	Monitoring Mechanisms	Assumptions and Risks
4. Improved 247 km of national highways of regional significance	Completed by June 2008 at appropriate quality and cost		
Activities 1. Institutional reform program		Proposal by Balochistan Provincial Government (BPG)	Capacity to implement system exits.
a. Strengthen planning and budget capacity with delegation of tasks and separation from executive functions	Implemented 2003-2005		Commitment to reform and institutional changes continues.
b. Expand road asset management system at CWD	Jan 2004 – Dec 2007	Proposal by BPG	
c. Develop new maintenance procedures, revise maintenance manuals, and prepare contracts	Jan 2004 – Dec 2007	Progress reports	Institutional capacity exists.
d. Analyze maintenance funding requirements and design a charging and a funding mechanism for road asset preservation	Jan 2004 – Dec 2006		Rules are enforced.
e. Build capacity at Road Environment and Social Assessment Cell (RESAC) at CWD	Appointment of three core staff by end 2003 and conducted training	Consultant report CWD proposal with action plan	
f. Implement axle-load control program with awareness and information campaigns	Jan 2004 – Dec 2006	Progress reports and reviews	
g. Implement provincial road safety action plan	Jan 2004 – Dec 2006	Progress reports and reviews	Road safety measures are effective.
h. Increase private sector share in periodic and routine road maintenance	10% contracted out in 2005 20% contracted out in 2006		Local contractors responded. Risks are appropriately shared.
2. Implement provincial road component	Jan 2004 – Dec 2008	BPG proposal and confirmation	
3. Implement action plan for improved cross-border movements	Implemented 2004-2006		
4. Develop, design, and construct a new cross-border facility at Chamman with improved trade	Implemented 2004-2006	Progress reports and reviews	Advanced procurement action results in speedy preconstruction activities, with timely award of contracts.
5. Implement national highway component.		Progress reports and reviews	
Inputs 1. Project funding total $267.3 million: ADB, $185.7 million, 69.5%; Government of Pakistan, $81.6 million (30.5%)	Loan approval in Oct 2003; and effectiveness in Jan 2004	Loan Agreement signed on time Annual financial statement	Counterpart funds are adequate and available on time.
2. Consulting services			Advance action results in speedy recruitment of consultants.
a. Design and construction supervision, preparation of subprojects, and benefit monitoring for CWD component; institutional development and capacity building at CWD	Jan 2004–Jun 2009	ADB's loan ledger Progress reports Subproject approval requests and approvals	All subproject preparatory and implementation work is done in a timely manner.
b. Design and construction supervision, and benefit monitoring of NHA component	Jan 2004–Jun 2008	Periodic review missions	International contractors responded. Local contractors perform well.
3. Civil works contracts			
a. National highways	Jan 2005–Jun 2009		
b. Provincial roads	Jan 2005–Jun 2009		

PAVED ROAD NETWORK IN BALOCHISTAN[a]

(km)

District	Roads			
	Access	Secondary	Highways	Total
Awaran	40.0	86.0	409.0	1309.0
Barkhan	168.0	143.9	112.9	424.8
Bolan	131.8	172.1	190.2	494.1
Chaghai	102.4	370.0	525.3	944.7
Dera Bugti	75.0	229.0	403.2	707.2
Gawadar	320.0	47.0	629.0	996.0
Jaffarabad	157.1	64.0	126.0	347.2
Jhal Magsi	63.3	134.1	258.6	456.1
Kalat	173.0	895.8	141.6	1,210.4
Kharan		526.0	791.0	1,317.0
Khuzdar	403.0	738.0	253.0	1,394.0
Killa Abdullah	115.0	445.1	215.6	775.7
Killa Saifullah	92.0	625.0	227.0	994.0
Kohlu Agency	52.0	257.0	534.0	843.0
Lasbela	87.6	302.0	8.3	397.9
Loralai	32.0	106.0	314.9	452.9
Mastung	23.2	334.1	12.4	369.8
Musakhail	106.0	117.0	99.0	322.0
Naseerabad	67.2	119.0	76.6	262.8
Panjgur	282.6	627.6	606.4	1,516.6
Pishin	124.2	313.7	287.4	752.3
Pishin / Quetta	237.9	14.4	0.0	252.3
Quetta	2.5	141.4	182.5	326.4
Sibi	48.0	68.8	230.8	347.6
Turbat	335.0	507.0	616.0	1,458.0
Uthal	478.1	34.0	286.0	798.1
Zhob	122.5	339.5	470.0	932.0
Ziarat	101.1	181.0	139.4	421.5
Grand Total	**3,940.4**	**8,659.5**	**8,146.3**	**20,746.2**

[a] Excluding national highways administered by National Highway Authority.

Source: Communication and Works Department, Balochistan.

ROAD SECTOR ANALYSIS

A. Overall

1. Transport is an important sector in Pakistan, contributing about 10% to the total gross domestic product (GDP). Road transport is the dominant mode of transport, and carries 90% of all passenger traffic and 95% of all freight traffic. Road transport services are largely in the private sector[1] and are subject to strong competition. During the 1990s, the overall demand for road transport grew at 7–8% per year while the average GDP grew at 4–5%. The number of registered motor vehicles is 4.2 million and is growing at 8% annually. Pakistan's national road network extends some 250,000 kilometers (km)? of which 60% is paved? and has grown about 4.5% annually during the last decade.

2. A total of PRs207 billion was raised in revenues from the road sector through a combination of general revenue taxes and user charges over the 5-year period 1995-2000. National and provincial highway expenditures totaled PRs118 billion over that period. The balance? PRs89 billion (43%)? was contributed to the government's general revenue.

B. National Highway System

3. **Network.** The national highway (NH) network comprises about 8,500 km of national highways and motorways,[2] and carries 75-80% of Pakistan's commercial traffic. The main domestic artery is concentrated along the Karachi–Lahore–Peshawar corridor, which is 1,670 km long. It serves 80% of Pakistan's urban population and carries over 55% of the country's intercity traffic. A detailed condition survey carried out in 2001 indicates that 50% of the NH network is in poor to very poor condition.[3] Some of the poor sections are located on the main international arteries (NH5 and NH25), connecting Pakistan with Afghanistan and providing access to ports in southern Pakistan.

4. **National Highway Authority (NHA).** NHA, with responsibility for operation, maintenance, and development of the NH system, is an autonomous highway agency with a functional structure and decentralized implementation through regional offices at provincial levels. The Asian Development Bank (ADB) and the World Bank have provided assistance[4] to reorganize and commercialize NHA, shifting its focus toward a network operator and service provider role and strengthening NHA's financial management systems to improve governance. NHA has, however, internal capacity constraints, such as limited in-house capability in strategic planning, project and asset management, and network operation. The use of modern road asset management and maintenance systems (including HDM-4) for operation and maintenance need to be fully institutionalized and integrated in the day-to-day operations.[5] Until recently, the four provincial highway departments were responsible for some 90,000 km of roads. Some parts of that network are now being transferred to the district administrations as part of the devolution program.

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1 About 95% of all road freight transport is carried out by the private sector.
2 The National Highway Authority (NHA) under the Ministry of Communications is responsible for the national highway network.
3 40% good, 11% fair and 49% poor/very poor.
4 ADB. 2001. *Report and Recommendation of the President to the Board of Directors on Proposed Loans to the Islamic Republic of Pakistan for the Road Development Sector Program* (program loan component). Manila.
5 Most of these issues are being addressed under a national road project, which is being prepared by the World Bank.

5. **Project Prioritization and Portfolio Management.** In the past NHA suffered from poorly justified investments, began too many new projects, and did not pay sufficient attention to network conservation. The situation led to the deterioration of road conditions along some of the more heavily trafficked highway sections, caused project delays, increased project costs, and reduce the headroom for new high-priority initiatives as well as conservation of the network. Encouraged by ADB and the World Bank, NHA has during the last 2 years reprioritized and reprogrammed the investment program with more attention and resources for conserving the NH network. Thus, the federal Government and NHA have initiated a multiyear PRs35 billion National Highway Improvement Program covering rehabilitation, resurfacing, and improvement of some 2,700 km of the NH network, based on a condition survey and a network-level prioritization study.

6. **Vehicle Overloading.** Vehicle overloading is a major impediment to sustainable development of the road sector in Pakistan, particularly on the NH network. Surveys by road agencies indicate that 88% of all trucks exceed the per-axle limit of 8 tons; 43% exceed the new proposed federal load limit of 12 tons per axle. Overloading causes roads to deteriorate rapidly, reducing their service life and significantly reducing the economic benefits that should have been realized during the designed life of the roads.[6] Moreover, the high recurrent costs of maintenance and the need for reconstruction erode the limited development resources of the country. As a result, there is a great need to strengthen enforcement of rules against overloading vehicles and accelerate the ongoing nationwide axle load program pursued at the federal level by the NHA.[7]

7. **Road Maintenance**. At NHA, it recognized increasingly that greater emphasis and more spending on road maintenance are necessary to preserve the national road infrastructure asset base. A road fund was established in 2001 with World Bank support. NHA has increased its spending on routine and periodic road maintenance; spending on routine maintenance increased from PRs600 million in FY1997/98 to PRs800 in FY2002/03, but further increases are necessary to reduce the backlog of maintenance. The increase of the road asset base, however, indicates that NHA should spend at least about PRs3,000 billion to keep the network in its present condition. Maintenance is increasingly being contracted out to private civil works contractors.

8. **Road Safety.** The fatality rate on Pakistan's road network remains among the highest in Asia.[8] The causes are high speeds, lack of road safety measures, poor discipline and enforcement of rules, and poor condition of vehicles. Road safety audits are not conducted routinely for all new and improved roads. The road safety problem is exacerbated by lack of awareness and poor knowledge of road safety matters among the general public and the road users. Efforts and coordination at the national and province levels are not concerted, and responsibilities and accountability among the institutions and provincial departments are unclear. Extensive efforts in education and training of drivers, coupled with awareness building, are required to address the high fatality rate. The Government passed the National Highway Safety Ordinance in September 2000, which provided the legal basis for establishing the National Highway and Motorway Police (NHMP) Force under the federal Ministry of

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[6] Calculations for local conditions show that a truck loaded at 12 tons per axle destroys roads five times as fast as a truck loaded at 8 tons per axle.

[7] The nationwide axle-load program pursued by the NHA includes installing 30 weigh bridges at strategic locations on the national highway network.

[8] Pakistan had 18.69 fatalities per 10,000 vehicles in 1992. This was lower than those for India (25.27) and Bangladesh (85.58), but higher than those for Southeast Asia (5-15 fatalities per 10,000 vehicles) and far above the countries with the lowest number of fatalities, such as the United Kingdom (1.60) and Japan (1.74). Source: ADB. 1995. *Regional Initiatives in Road Safety*. Manila.

Communications. This measure expands the role of the very successful motorway police to the national highway system, in a phased manner.

9. **Private Sector Participation**. NHA is pursuing opportunities for private sector particupation (PSP) in road infrastructure projects. Projects include the Lahore and Karachi bypass projects, which are being prepared on build-operate-transfer (BOT) basis. The role of the domestic private sector in road contracting and providing engineering services is fairly well developed. NHA needs, however, to step up its efforts to increase PSP. NHA is currently developing Lakhpass crossing just outside Quetta in Balochistan on BOT basis. NHA could increase PSP manifold by introducing network management and performance-based contracts for maintenance.

C. Balochistan Province

10. **Institutional Capacity and Constraints.** The Communication and Works Department (CWD) in Balochistan, which is responsible for roads and buildings, is structured and functions as a typical public works department. The key issue at CWD is inadequate funding for maintenance and weak institutional capacity in planning and executing maintenance. Thus, there is a great need to build institutional capacity for maintenance and revamp the road asset management system. Although a road maintenance unit was established in 1999 with support of ADB and the Nordic Development Fund, maintenance did not become fully institutionalized and integrated in the day-to-day operations at CWD. Another important but more difficult issue is rightsizing of staff. This needs to be addressed with a broader institutional and organizational review of CWD and a move toward a modern road management agency. Moreover, as a consequence of the ongoing devolution program,[9] which was formally initiated in August 2001, the functions and allocation of responsibilities between CWD and the 26 district administrations need to be reviewed.

11. **Road Maintenance.** Since 1995, spending on road maintenance in Balochistan has remained largely constant; budget allocations for road maintenance[10] increased by 4.9% from PRs92.3 million in FY1998 to PRs97.8 million in FY2002, while the allocations for road development (investments) under the annual development grant were around PRs2.121 billion in FY2002 and PRs3.502 billion in FY2003. The recent expansion of the road network and the increase of the road asset base indicate that further increases of budget allocations for maintenance are required to sustain the road network.[11] While Balochistan has lower road density and generally poorer access than other provinces in Pakistan, some road network expansion is still required to connect districts and villages, and thus provide access for the poor rural communities. It is also important to increase maintenance spending and improve maintenance management and processes on existing roads.

12. **Road Safety.** Improving road safety is also an important issue in Balochistan. The situation is exacerbated because of weak institutional capacity at CWD and the narrow roads and poor conditions. Under the Project, CWD will prepare an action plan to address road safety with preventive and corrective measures including information and awareness campaigns and education and adoption of road safety audits.

13. **Private Sector Participation**. The opportunities for PSP in road infrastructure projects are smaller in Balochistan than elsewhere in Pakistan due to remoteness and relatively low traffic volumes. The role of the domestic private sector in road contracting and providing

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[9] Under the devolution program, local governments at distric level have been created.

[10] Road maintenance spending level in Balochistan: FY1998/99? PRs77.3 million. FY1999/2000? PRs47.5 million. FY2000/01? PRs77.4 million. FY2001/02? PRs 100.9 million.

[11] Based on analysis by consultants supporting the road maintenance unit at Communication and Works Department.

engineering services is, however, fairly well developed. Both NHA and BPG need, however, to step up efforts to increase PSP in Balochistan. NHA is currently developing Lakhpass crossing just outside Quetta in Balochistan on BOT basis. CWD could increase PSP by contracting out maintenance and gradually introducing network management and performance-based contracts for maintenance.

EXTERNAL FINANCING FOR THE ROAD SECTOR

Project	Funding Source	Amount ($ million)	Year
Farm-to-Market Roads Project	ADB	38.00	1986
Second Farm to Market Roads Project	ADB	113.00	1990
Flood Damage Restoration Project	ADB	78.00	1993
Sukkur Bridge Project	ADB	45.00	1994
Rural Access Roads Project	ADB	123.00	1996
Road Sector Development Program	ADB	200.00	2001
Punjab Road Development Sector Project	ADB	150.00	2002
Subtotal		**747.00**	
Locomotive Manufacturing Factory	JBIC	81.00	1984
Railway Transportation Reinforcement (II)	JBIC	123.00	1985
Indus Highways - Phase I	JBIC	71.00	1989
Indus Highways - Phase II	JBIC	173.00	1991
Track Circuit at 94 Mainline Station	JBIC	27.00	1992
Rural Roads Construction	JBIC	95.00	1993
New Locomotive Production	JBIC	50.00	1993
Indus Highways - Phase IIB	JBIC	152.00	1993
Diesel Electric Locomotive Rehabilitation	JBIC	50.00	1993
Kohat Tunnel Construction (I)	JBIC	45.00	1994
Diesel Locomotive Production II	JBIC	71.00	1996
Diesel Electric Locomotive Rehabilitation	JBIC	56.00	1996
Survey Vessel Modernization	JBIC	20.00	1996
Kohat Tunnel Construction (II)	JBIC	34.00	2001
Subtotal		**1,048.00**	
First Highway Project	WB	17.00	1964
Third Highway Project	WB	50.00	1980
Fourth Highway Project	WB	152.00	1987
Transport Sector Project	WB	184.00	1991
Karachi Port Modernization	WB	91.40	1992
Trade and Transport Facilitation	WB	3.00	2001
Subtotal		**497.40**	
	Total	**2,292.40**	

ADB = Asian Development Bank, JBIC = Japan Bank for International Cooperation, WB = World Bank.
Source: Asian Development Bank estimates.

PROVINCIAL POLICY MATRIX AND ACTION PLAN

Main Areas of Reform	Objective	Key Issues and Actions Taken	Next Step of Reform and Actions	Compliance Monitoring
Institutional Reform with New Organization Structure	Transform the organizational structure of Communication and Works Department (CWD) so it becomes an effective and modern road management organization	The organization of the CWD went through several changes recently. Initially, many functions were transferred to the districts. This action did not prove successful. A new organization structure was approved on 4 March 2003, which brought back some functions to the provincial level. Rightsizing of staff is a key issue in the move toward a modern road management agency.	Continue steps to transform CWD so it becomes a modern road management agency focusing on provincial network modernization and conservation, with increased emphasis on maintenance. Review the organizational setup and reorganize CWD along modern road management principles by June 2005 and approve new setup end-2005. Delegate appropriate road management tasks, specifically maintenance, to district administrations in support of devolution. Consider converting CWD into a separate highway authority or corporation.	Prior to approval of the new institutional setup and organizational structure and staffing plan, CWD will submit the proposal and the implementation plan to the Asian Develoment Bank (ADB) for comments.
Modern Maintenance Practices	Ensure that CWD has systems, procedures, and capacity to carry out effective and efficient road and bridge maintenance management	A road management unit (RMU) was established in 1998 with support of ADB technical assistance (TA) but it has since lapsed. The RMU needs to be empowered at CWD as the focal point for planning and monitoring maintenance. CWD needs capacity building, and training. Maintenance is at present rules -driven, labor-based, and performed by force account. Condition surveys are not institutionalized at CWD or the district level.	CWD, with the support of the consultants under the provincial institutional development component (PIDC), will develop road asset management system (RAMS), new maintenance procedures and manuals; and training staff in procedures for developing, procuring, and supervising contracts for routine and periodic maintenance based on competitive tendering. The new maintenance contracts will (i) include standard or normal contracting out and network management, and (ii) be performance-based.	Consultant's report CWD will submit the implementation plan with brief description of the new system and procedures.
Road User Charges	Ensure that funding of maintenance for long-term preservation of the provincial road network is adequate and sustained	Funding for maintenance is currently provided from the consolidated revenue account. This system is working satisfactorily. The main problem is the need to pay staff formerly employed in maintaining roads that have been transferred to the National Highway Authority (NHA).	CWD will, with support of consultants under the PIDC, analyze maintenance funding requirements and available options for secured and sustained funding for road asset preservation, specifically (i) a new road user charge to recover costs for maintenance, and (ii) the use of tolls on routes used by heavy through traffic. The preferred schemes for (i) road user charges, (ii) toll collection, and (iii) maintenance funding mechanism will be implemented from 2005 following consultations with ADB. .	Consultant's report CWD will submit the new proposal and implementation plan to ADB for comments.

Main Areas of Reform	Objective	Key Issues and Actions Taken	Next Step of Reform and Actions	Compliance Monitoring
Axle Load Control	Reduce overloading of vehicles, especially trucks to prevent early deterioration of the provincial road network	Overloading of trucks is a major and common problem in Pakistan. NHA is implementing a program for axle-load control for the national highway network. Only two have been built in Balochistan. Balochistan provincial government (BPG) is monitoring the effectiveness of these stations.	As part of the Project, weigh stations will be installed for selective provincial highways at strategic locations. Capacity and enforcement of rules will be strengthened through training of CWD and other staff, including the provincial police. Awareness campaigns will be carried out by CWD. CWD will prepare an action plan for improved enforcement of regulations, which will also link with the devolution and police reforms supported under ADB assistance.	CWD will submit an action plan for improved enforcement of regulations. Project progress reports Physical inspection of installed weigh stations
Road Safety	Improve road safety and reduce the total number of road accidents in the province of Balochistan	There is lack of coordination and capacity in BGP? specifically at CWD, education and health departments, and the police force? for preventive and corrective road safety measures including information campaigns, training, and education. No accident reporting centers have been established in Balochistan.	The objective is to develop, prepare, and implement a road safety action plan in Balochistan, comprising, inter alia, awareness campaigns, strengthening of education in schools, drivers' training, and enforcement of traffic rules. The PIDC will support the committee's work and CWD will, by 2006, institutionalize road safety audits for all road projects.	Road safety action plans will be submitted to ADB for comments by end-2005. Status reports on action taken to improve enforcement of road safety and traffic regulations will be made.
Capacity Building for Environment and Social Assessment	Strengthen CWD's institutional capacity for environment and social assessment of road projects	CWD lacks capacity and experienced staff for environmental and social assessment. There is a great need to strengthen monitoring of civil works contracts compliance with labor laws, specifically gender inequality and child labor.	CWD will be strengthened through capacity building and training of more staff, to be able to carry out environment and social assessments of road projects. Capacity building will be provided under the PIDC.	Project progress reporting Project review missions
Private Sector Involvement	Increase private sector participation, wherever feasible, in CWD's road maintenance operations and road rehabilitation program	The opportunities for PSP in road infrastructure projects have been smaller in Balochistan than elsewhere in Pakistan due to the remoteness and relatively low traffic volumes. Use of contractors for maintenance is relatively undeveloped in Balochistan.	CWD will increase contracting out of maintenance so that 10% of routine and periodic maintenance will be undertaken by the private sector in 2005, 20% in 2006, and 40% in 2007.	Yearly status report with completed actions will be submitted to ADB.
Road User Associations	Increase involvement and participation of the public and road users in road operations and development	A few road user associations are established in Pakistan and involved in various areas, such as education and road safety awareness campaigns.	BPG and CWD will encourage the establishment of road user associations and similar organizations for broader participation and more involvement by the public and road users in road operations and development in Balochistan.	Project progress reporting

SUMMARY OF REFORM PROGRAM OF NATIONAL HIGHWAY AUTHORITY

Policy Theme	Objective	Key Issues and Actions Taken (under Loan 1891-PAK)	Next Step of Reform and Actions	Status and Compliance Monitoring
Transport Policy	Establish a transport system that provides efficient, safe, reliable, equitable, and environmental-friendly access and mobility for people and goods.	The transport policy was cleared by interministerial review and finalized for submission to the Cabinet.	Further improve and strengthen transport policy and develop performance indicators Adopt new transport policy by 30 June 2005 and subsequent approval by Cabinet	Government Notification Project review missions
Road Sector Resource Management	Invest primarily in roads that are economically viable and for which stable and secure resources for preservation of assets are available	A medium-term (5 year) sector investment plan has been prepared by National Highway Authority (NHA) that prioritizes projects on the basis of economic returns. The plan focuses on ongoing priority projects, appropriately balancing network expansion and conservation.	Allocation of road sector revenues for road development and maintenance follow the medium-term plan. The medium-term sector investment plan will be reviewed annually to ensure that it meets the emerging priorities.	Project review missions
	Improve road maintenance and increase allocations for road maintenance	NHA has increased road maintenance allocation to PRs2.5 billion in FY2001/02 and further to PRs3.4 billion in FY2002/03.	Road maintenance allocations will be further inceased to meet needs of the entire national road network by 2005.	Project review mission
	Implement a road asset management system (RAMS) to ensure that NHA effectively carry out asset management and preservation. Institutionalize the use of the road asset management system	RAMS is used as the basis for allocating maintenance expenditures. The Road Asset Management Directorate (RAMD) at NHA is fully functioning, with core staff (11 officers and 20 staff) posted at RAMD.	RAMS will be expanded and capacity of the staff enhanced. Adequate resources will be provided to undertake regular road condition surveys.	Project review missions
Road Asset Preservation	Introduce an axle load control program to protect and increase the service life of road assets	The Government has prepared and approved legislation. NHA has initiated actions to enforce legislation through awareness programs and measurement of vehicle load levels. Thirty high-speed weigh stations are operational, delayed but underway.	Implementation of axle-load control program will continue and coverage will be expanded. Provinces will be assisted to control vehicle overloading.	Project review mission
	Improve enforcement of axle load control and road safety	(MOCR) has extended the jurisdiction of the National Highway and Motorway Police (NHMP) for enforcing road safety and axle load control to the national highways, starting with 320 km of N5. A fine collection system has been set up and is operational at weigh stations for five national highways. Enforcement procedures and institutional framework have been	MOCR will continue to extend NHMP jurisdiction until full coverage over a 5-year period. Provinces will be assisted in establishing similar arrangements.	Project review missions

Policy Theme	Objective	Key Issues and Actions Taken (under Loan 1891-PAK)	Next Step of Reform and Actions	Status and Compliance Monitoring
		established including schedule of penalties for violation of load levels.		
Improvement of Institutional Efficiency	Operate road agencies on the principles of economy and efficiency in operations—move to a commercial road agency, i.e., operating on a fee-for-service basis	NHA has reorganized its structure and improved operational, management and financial performance. NHA has rightsized its staff strength from 1,670 to 1,219, which represents a 22% reduction in officer cadre and 28% reduction in staff below Grade 16.	Institutional evaluation exercise will be continued to create balance between the functions and staffing needs and achieve maximum efficiency. The current proposal for NHA reorganization is included..	Project review missions
Financial Management	Improve financial management.	NHA has introduced sound financial management practices that ensure proper accountability for use of funds. NHA has revised its standard operating procedures (SOPs) and its financial manual.	Implementation of SOPs and its financial manual will be initiated.	Project review missions
Road Safety	Introduce measures to reduce the unacceptably high toll of deaths and accidents on roads in Pakistan	The Government promulgated the National Highway Safety Ordinance in September 2000. The National Highway Safety Committee (NHSC) was established in 2001 for road safety and implementation coordination on national highways. Agencies was established. NHA and provincial communication works departments (CWDs) are the responsible agencies. Secretariats are being established and road safety plans are being prepared.	Enforcement, monitoring, and evaluation procedures for road safety initiatives will be strengthened. NHSC effectiveness will be improved and adequate resources will be provided. Activities will be expanded to provinces. NHA will continue its comprehensive nationwide stakeholder consultation process with regular biannual meetings.	Project review missions

DETAILED COST ESTIMATES

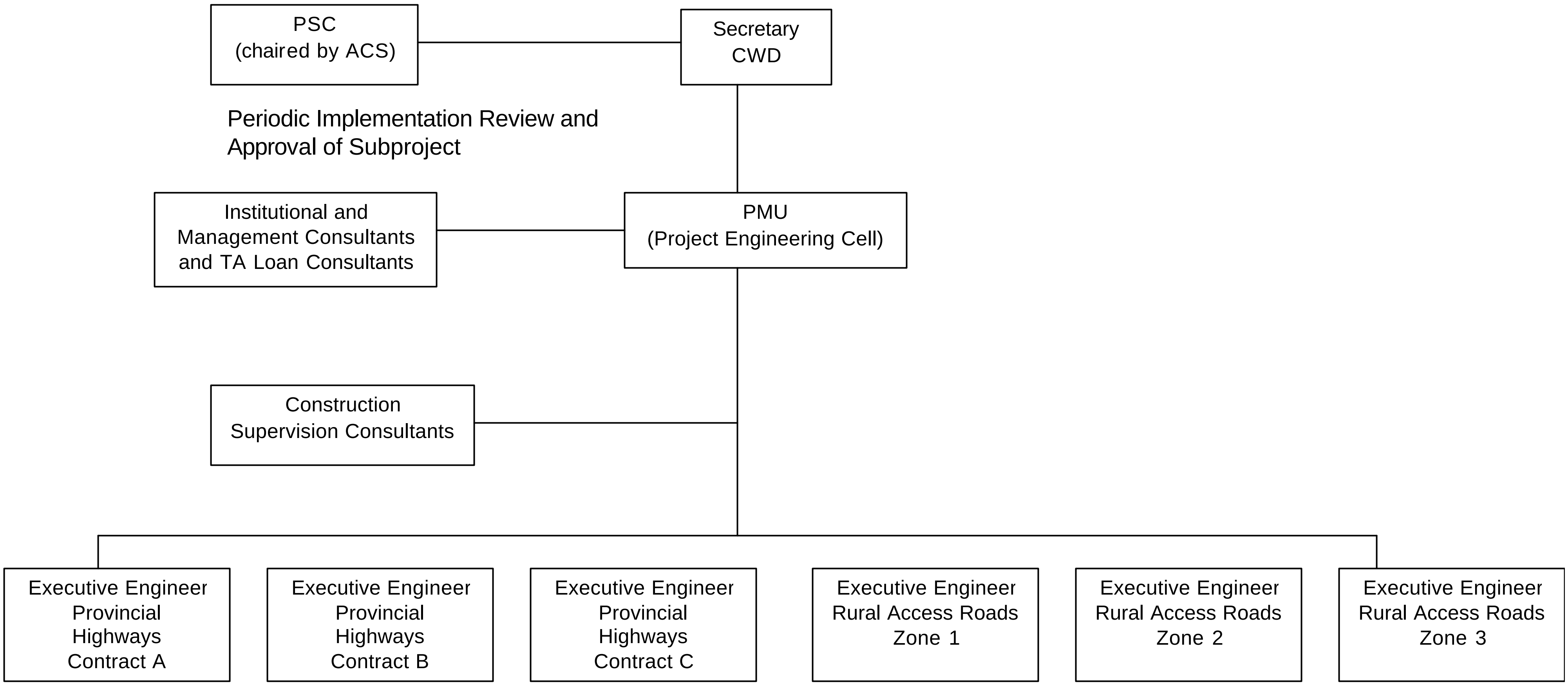
($ million)

Project Component		FX	Local	Total	% Total Base Costs
A.	**Provincial Institutional Development Component (PIDC)**				
1.	Consulting Services	1.60	0.53	2.13	0.90
2.	Equipment	1.50	0.50	2.00	0.85
	Subtotal (A)	**3.10**	**1.03**	**4.13**	1.75
B.	**Provincial and Rural Roads Improvement**				
1.	Land Acquisition and Resettlement	0.00	3.00	3.00	1.27
2.	Civil Works - Core Provincial Highways				
	a. Zhob - Mir Ali Khel - Khajuri Kach	8.89	5.92	14.81	6.28
	b. Nag - Panjgur	5.35	3.56	8.91	3.78
	c. Kach - Shahrag - Harnai	10.60	7.06	17.66	7.48
3.	Civil Works - NonCore Provincial Highways	11.62	7.74	19.36	8.21
	Subtotal (Provincial Highways)	**36.44**	**24.30**	**60.74**	**25.74**
4.	Civil Works - Core Rural Access Roads				
	a. Gulistan-Xagi Bazar-Kandel	0.53	0.35	0.88	0.37
	b. Dera Allahyar - Hardin	3.64	2.43	6.07	2.57
	c. Khurkhera - Kanraj	1.42	0.94	2.36	1.00
5.	Civil Works - NonCore Rural Access Roads	28.70	18.60	47.30	20.05
	Subtotal (Rural Access Roads)	**34.29**	**22.32**	**56.61**	
	Subtotal (Provincial Roads)	**70.73**	**49.62**	**120.35**	
6.	Maintenance Subcomponent	0.30	0.20	0.50	
7.	Consulting Services for Design & Supervision Adjustment	4.70	2.69	7.39	3.13
8.	Incremental Administration Costs	0.00	2.00	2.00	0.85
	Subtotal (B)	**75.73**	**54.51**	**130.24**	**55.20**
C.	**Cross-Border Development Component (CBC)**				
1.	Civil Works	1.80	0.70	2.50	
2.	Equipment	1.00	0.50	1.50	0.64
	Subtotal (C)	**2.80**	**1.20**	**4.00**	**1.70**
D.	**National Highways Component (NHC)**				
1.	Land Acquisition and Resettlement	0.00	1.00	1.00	0.42
2.	Civil Works				
	a. Quetta - Kalat	23.96	15.98	39.94	16.93
	b. Quetta - Chamman	30.75	20.50	51.25	21.72
3.	Consulting Services for Construction Supervision	1.53	1.86	3.39	1.44
4.	Incremental Administration Cost	0.00	2.00	2.00	0.85
	Subtotal (D)	**56.24**	**41.34**	**97.58**	**41.36**
	Total Baseline Costs	**137.87**	**98.08**	**235.95**	100.00
	Physical Contingencies	8.26	3.54	11.80	
	Price Contingencies	7.43	3.19	10.62	
	Total Baseline Costs Plus Contingencies	**153.56**	**104.80**	**258.37**	
	Front-End Fee	**0.93**		**0.93**	
	Interest During Construction	8.00	0.00	8.00	
	Total Project Cost	**162.49**	**104.80**	**267.30**	

FX = foreign exchange.

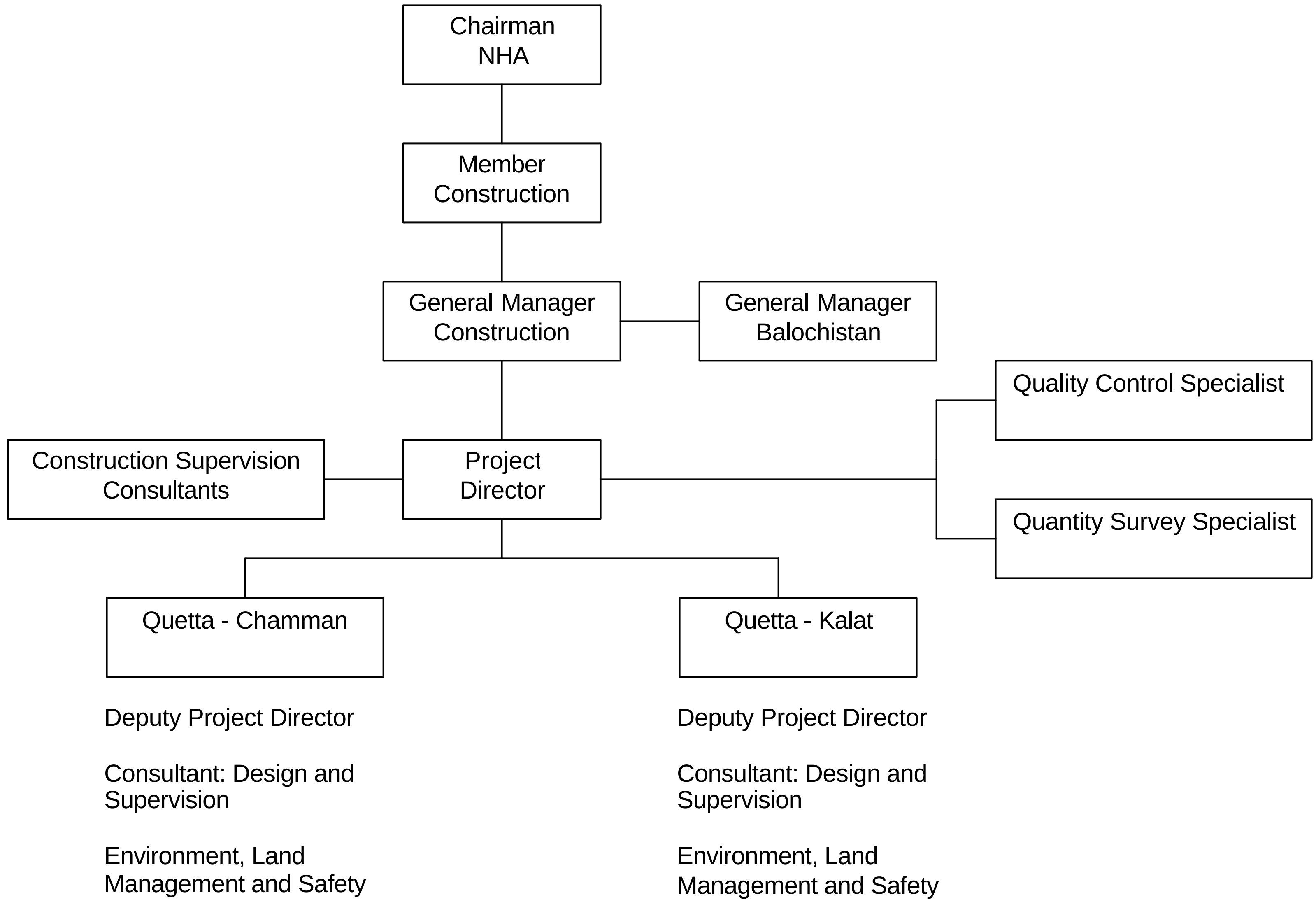
Source: TA 3897-PAK and Asian Development Bank staff estimates.

IMPLEMENTATION ARRANGEMENTS FOR PROVINCIAL COMPONENTS



ACS = assistant chief secretary, CWD = Communication and Works Department, PMU = Project Management Unit, PSC = project steering committee.

IMPLEMENTATION ARRANGEMENTS FOR NATIONAL HIGHWAY COMPONENTS



NHA = National Highway Authority.

SELECTION CRITERIA FOR PROVINCIAL ROADS

1. The selection of subprojects by the Balochistan provincial government (BPG) will be subject to Asian Development Bank (ADB) approval. The criteria for selection and the approval process are outlined in this appendix.

A. Subproject Selection Criteria

2. Subprojects must meet the following criteria.

(i) **Provincial road master plan.** The subproject will be part of the provincial road master plan.

(ii) **Poverty targeting**. The subproject will have been prioritized based on the district development ranking used by the Government for determining allocation of its development expenditures.

(iii) **Technical feasibility**. The subproject will be technically feasible and a detailed report prepared.

(iv) **Economic feasibility**. Economic analysis of the subproject will be prepared in accordance with ADB's *Guidelines for the Economic Analysis of Projects* and the estimated economic internal rate of return will be at least 12%.

(v) **Social assessment and analysis.** The subproject will be socially sound, will minimize the need for land acquisition, and will include measures to mitigate any social impacts. An initial social assessment (ISA) for the subproject will have been prepared in accordance with ADB's relevant guidelines.

(vi) **Resettlement.** The proposed subproject road will not entail any involuntary resettlement or land acquisition. Impact on any affected persons through land acquisition or description of livelihood will be minimal. If a subproject involves land acquisition or involuntary resettlement, a full resettlement plan (if the number of affected persons is 200 or more) or a short one (if fewer than 200 persons are affected) acceptable to ADB, will have been prepared in accordance with the resettlement framework agreed upon by the Government and ADB and in accordance with ADB's *Guidelines on Involuntary Resettlement*.

(vii) **Indigenous people.** If any indigenous people, such as tribes or settlers, as per ADB's definition, are likely to be affected significantly by a subproject, an indigenous people's plan should be prepared in accordance with ADB's *Policy on Indigenous People*.

(viii) **Environmental considerations**. Each proposed subproject road will have been screened for environmental impacts in compliance with the requirements of the Government and the ADB,[1] and should not cause any major adverse impact on the environment. ADB will have received and reviewed the initial environmental examination, if any, prepared for the proposed road. All necessary approvals and clearances will have been obtained for the subproject from the federal and provincial governments and related agencies.

(ix) **Counterpart funding.** BPG will ensure that sufficient counterpart funding is allocated to implement the subproject on time and efficiently in accordance with the overall implementation schedule.

(x) **Government approvals.** All necessary BPG approvals will have been

[1] ADB. 2003. *Environmental Assessment Guidelines of the Asian Development Bank*. Manila.

obtained for each subproject.

(xi) **District commitment.** As appropriate, BPG will have confirmed, in a manner acceptable to ADB, its commitment of the districts to the proposed subproject road, including but not limited to the provision of budgetary allocations for rehabilitation and operation and maintenance for the subproject road.

B. Approval Process for Subprojects

3. All subprojects will be required to comply with ADB's policies and guidelines and satisfy ADB's procedures for subproject preparation with respect to technical, operational, economic, social, indigenous people, resettlement and other requirements. BPG will ensure that sufficient counterpart funding will be available. The six core subproject provincial roads for which the aforementioned requirements were applied during project preparation were found technically feasible, economically viable, socially responsible and environmentally sound. For additional subprojects, the requirements will be applied during project implementation.

4. During the early stages of implementation, the project management unit (PMU) through the Project Engineering Cell an with the assistance of the institutional and management consultants, will prepare a consolidated report for one or more subprojects covering all the stipulated ADB requirements. The report will be sent to ADB for review and approval. With ADB's approval, PMU will proceed with further subproject preparation, procurement, and subproject implementation.

IMPLEMENTATION SCHEDULE

Project Component	2003		2004				2005				2006				2007				2008				2009			
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Consulting Services for CWD																										
Design and Supervision Consultant Selection		■	■																							
Detailed Design of Balance Roads				■	■	■	■	■	■	■	■															
Construction Supervision				■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		
Institutional Development and Management:																										
Consultants Selection				■	■	■																				
Institutional Development Consultant Work							■	■	■	■	■	■	■	■	■	■	■	■								
Construction of Provincial Roads																										
1st 400 km of Provincial Roads																										
Contractors Prequalification		■	■																							
Bidding and Award of Contract				■	■																					
Construction						■	■	■	■	■	■	■	■	■	■	■	■	■								
2nd Phase of Provincial Roads																										
Bidding and Award of Contract								■	■																	
Construction										■	■	■	■	■	■	■	■	■	■	■						
3rd Phase of Provincial Roads																										
Bidding and Award of Contract												■	■	■												
Construction															■	■	■	■	■	■	■	■	■	■		
Consulting Services for NHA																										
Consultant Selection		■	■																							
Design, Review, Update Bid Documents				■	■																					
Bid Evaluation						■																				
Construction Supervision							■	■	■	■	■	■	■	■	■	■	■	■	■							
Construction of National Highways																										
Contractors Prequalification			■	■																						
Quetta–Kalat Section																										
Bidding and Award of Contract					■	■																				
Construction							■	■	■	■	■	■	■	■	■	■	■	■	■							
Queta–Chamman Section																										
Bidding and Award of Contract						■	■																			
Construction								■	■	■	■	■	■	■	■	■	■	■	■	■	■					
Cross-border Facility Construction											■															

CWD = Communications and Works Department, NHA = National Highway Authority.

INDICATIVE CONTRACT PACKAGES

Package No.	Road Section	Length (km)	Completion Time (months)	$ million
National Highways				
ICB 1	Quetta Kalat Road (Section I)	70	36	21.5
ICB 2	Quetta Kalat Road (Section II)	60	36	18.4
ICB 3	Quetta–Chamman (Section I)	60	36	26.3
ICB 4	Quetta–Chamman (Section II)	57	36	25.0
Provincial Roads				
ICB 5	Zhob–Mir Ali Khel–Khajuri Kuch	101	36	14.8
ICB 6	Nag Panjgur	85	36	8.9
ICB 7	Kach–Shahrag–Harmai	95	36	17.7
LCB 1	Gulistan Sagi Bazaar Kandel	13	18	0.9
LCB 2	Khurkhera Kanraj	24	24	2.4
LCB 3	Dera Allah Yar–Hairdin [a]	20	24	3.0
LCB 4	Dera Allah Yar–Hairdin [a]	20	24	3.0

ICB = international competitive bidding, LCB = local competitive bidding, NHA = National Highway Authority.
[a] A 7.3 meter wide section.
Source: National Highway Authority and Balochistan Communication and Works Department.

TECHNICAL ASSISTANCE LOAN FOR

COMMUNITY DEVELOPMENT AND POVERTY REDUCTION

A. Background

1. Balochistan is a relatively poor province and economic growth is constrained by a road network suffering from lack of development and maintenance. It is the largest of Pakistan's provinces, but it has isolated settlements and is generally detached from the country's mainstream political and economic activities. The Project has been designed to support sustainable pro-poor growth, reduce poverty, promote subregional cooperation, and be environmentally and socially responsible. The Project is expected to contribute significantly to poverty reduction in the medium term by increasing literacy level, reducing transport costs, reducing intradistrict travel times, increasing access to social services, and reducing the vulnerability of the population, particularly of women. According to surveys conducted as part of project preparation, people along the roads expect the roads to generate employment opportunities for them and increase their access to social services like health and education. Similarly poor people would be able to reach police stations and banks more easily. The Project will help people transport their goods to market, reduce transport costs, increase food security, and increase mobility of the population.

2. Experience elsewhere has shown that the potential benefits to the poor are likely to be achieved quickly. The effects are more likely to be sustained if the infrastructure project is augmented with specific support and assistance targeted at the poor, such as direct involvement in road construction and construction of ancillary facilities (footpaths, drainage), or indirect involvement through activities that enable the community to capitalize on the new opportunities. The latter might include drilling wells to enable cash crops to be planted or providing transport to increase access to health and education. In both cases the emphasis is on providing training and skills to help the poor help themselves.

B. Objectives

3. The technical assistance (TA) will (i) develop and implement community initiatives that leverage off the Project in a manner that increases its positive impact on the poor, and in particular, on women; (ii) support intitiatives aimed at promoting greater awareness of social, health and safety issues that may emerge from the Project; and (iii) establish a scheme to measure and monitor the poverty impact of roads.

C. Major Activities and Tasks

4. The TA involves three major activities, which may overlap and complement each other.

5. **Community Infrastructure and Empowerment Fund.** Grants of up to the equivalent of $25,000 each in support of poverty alleviation in road-side communities along roads supported under BRDSP will be made on a competitive basis to selected non-governmental organizations and other community groups approved by ADB.

6. **Social Protection and Education Fund.** Grants up to the equivalent of $10,000 each in support of programs aimed at raising community awareness of social, health and safety issues such as prevention of HIV/AIDS, trafficking of women and children, and vehicular traffic accident will be made on a competitive basis to selected non-governmental organization and other community groups approved by ADB.

7. **Monitoring and Measuring Social and Poverty Impacts**.
   (i) Identify indicators for measuring the social impact of the project roads and associated programs.

(ii) Undertake a benchmark study to determine the current values of the measures, identify the current use of the road, and describe the socioeconomic profile of users and nonusers.
(iii) Training the Project management unit/Project Engineering Cell in the CWD to identify and monitor social impacts, and establish an ongoing monitoring program.
(iv) Prepare a baseline report, a project completion report, and a review report 2 years after physical project completion.
(v) Carry out a socioeconomic and poverty impact study on past road projects.

8. **Cost and Financing.** The TA will be financed from the Special Funds resources of the Asian Development Bank (ADB). The total cost is estimated at $1.25 million comprising $0.22 million in foreign exchange and $1.03 million in local currency. The TA loan of $1.0 million will finance the total foreign cost and $0.78 million of the local cost. The Balochistan provincial government (BPG) will finance the remainder of the local costs.

E. Implementation Arrangements

9. **Executing and Implementing Agencies.** CWD will be the Executing Agency. The established project management unit (PMU) at CWD will be the Implementing Agency, headed by the project director. Nongovenment organizations (NGOs) will be appointed to undertake the TA loan activities. The TA is planned to begin by January 2004 and will be implemented over six years.

10. CWD will in consultation with ADB, establish a Community Support Project Committee (the Committee) consisting of no less than seven members. At least four members of the Committee shall be drawn from non-governmental organizations, academia or the media. At least three members of the Committee shall be women. This committee will be entrusted to develop and approve detailed criteria and procedures for solicitation and selection of applications for grant awards under the (i) Community Infrastructure and Empowerment Fund and (ii) Social Protection and Education Fund. In setting criteria for grant selection, the Committee shall give utmost importance to enhancing livelihood opportunities of the poor, particularly women, and in addressing other basic needs concerns such as clean water supply and nutrition.

11. **Procurement of Goods and Services.** The PMU will initiate the hiring of consultants. Each consultant or consulting team will prepare an inception report for its subcomponent within 2 weeks of the start of services, providing a time-bound work program and a schedule of progress reports. For each subcomponent, a draft final report will be submitted and a final report finalized after comments of the Government and ADB have been incorporated.

12. **Training, Workshops, and Seminars.** Community participation is the essence of the TA. Workshops will be held on each road to introduce the Project and to provide a means of communicating concepts and reporting progress. The TA will train CWD and the community groups to ensure effective transfer of knowledge.

13. **Accounts and Audits.** CWD will establish separate project accounts for expenditures under the TA loan, with separate recording of transactions. CWD will appoint an independent auditor acceptable to ADB to routinely monitor and annually evaluate and document the execution of community initiatives supported under the Project. Such auditors will be replaced every two years.

TECHNICAL ASSISTANCE GRANT FOR

CROSS-BORDER DEVELOPMENT

A. Objective

1. The Technical Assistance (TA) will develop a concept, prepare a plan, and complete a package, including outline design and specifications, for a pilot cross-border facility to be established and operated at the Chamman border point. The pilot cross-border facility will take cognizance of the environment and opportunities for coordinated operation of immigration, customs, vehicle inspections, and health and other phytosanitary inspections. The facility must be administratively least cumbersome, client-friendly, and transparent in all processes while fully complying with applicable government policies, regulations, and interests.

B. Scope and Outline Terms of Reference

2. The TA will entail the following major tasks.

(i) Briefly review all significant recent (last 5 years) assessments of trade facilitation arrangements (including port clearance and storage fees and other levies) at ports and border crossings conducted by the Government of Pakistan or internal or external agencies, ascertain the validity of such assessments, and analyze why any proposals, if any, have not been implemented.

(ii) Review the major legal, financial, and administrative issues affecting cross-border transit and trade between Pakistan and Afghanistan, and identify any major gaps and required changes in the 1965 Transit Agreement to develop a comprehensive transit agreement that considers the interests and concerns of both governments and the competitive situation.

(iii) Identify, review and assess the key issues on transit of goods and vehicles carrying such goods. The review will include, but be will not be limited to (a) regulation of vehicle movement across the borders; (b) harmonization of vehicle standards on dimensions, weights, axle loads, emissions, etc.; (c) harmonization and cross-acceptability of roadworthiness inspection criteria; (d) development of a cross-border insurance regime; and (e) introduction of a customs temporary vehicle importation regime (i.e., waiver of import duties without posting any guarantee).

(iv) Evaluate the "one-stop" pilot trade facilitation arrangements at Qasim Port and Karachi Port and any border crossings where similar efforts may be underway, and delineate the primary causes of the successes and failures of such efforts. In addition, assess further potential for document alignment to make it responsive to the common requirements of buyers, sellers, banks, insurers, forwarders, port and customs authorities and any others that may be involved in the transit process.

(v) Map every required step for landing, unloading, warehousing, clearing and reloading goods at Karachi and Qasim ports and Chamman/Spin Boldak border crossing and analyze costs for each step in time and money (including any "facilitation fees" that may be charged by unauthorized agents); and make a detailed analysis of how each step could be improved, streamlined, rationalized, consolidated, or eliminated to gain efficiency.

(vi) Compare the mapped existing processes with the best practices in the region and elsewhere and detail what will be required to make the practices in Pakistan conform to best practices in the region and to adapt to the local environment.

(vii) To test the validity and applicability of the results of the above analysis, hold workshops and other consultations with (a) the established trade facilitation group managed by the Ministry of Commerce (MOC) and its members from the public and private sector; (b) users of such cross-border facilities, including operators of vehicles and owners of goods being shipped; and (c) current administrators and other operators of cross-border facilities.

(viii) Develop a plan of operations for the pilot cross-border facility including, but not limited to, preparation of rationalized procedures for compliance with all border clearance requirements (e.g., single-form processing) and staffing needs and training (by different identified categories of skills).

(ix) Coordinate closely and consult, as necessary, with government ministries and other agencies to develop a consensus on the cross-border concept, and plan and proposed facility? including systems and equipment? and the management responsibility for the pilot cross-border facility and how management oversight over representatives of multiple government ministries and agencies could be effected.

(x) Identify all needs for the cross-border facility and any related or additional facilities (including any warehousing, truck parking sheds, cargo inspection sheds, weigh bridges, rest facilities, etc.), equipment (including cargo scanning and sniffer dogs, telephone/Internet communications, firefighting equipment, etc.), and training to fully operate the pilot facility.

(xi) Identify other transit operations in Afghanistan and Pakistan with which communication links would be useful for effective operation of the pilot cross-border facility and development of a recommendation on how a communication network could be established among such transit operations.

(xii) Prepare guidelines for recommended organizational structures indicating number and type of personnel required.

(xiii) Identify the appropriate roles for the private and public sectors and investment opportunities.

(xiv) Prepare a manual of recommended procedures and available resources to improve cross-border operations and hold training workshops with all relevant staff working at the Chaman.

(xv) Prepare (a) a detailed concept plan, (b) a comprehensive investment package (including but not limited to a master plan, outline design, equipment and systems, and specifications for the new cross-border facility) for external and private sector financing, (c) a plan for a recommended organizational setup, (d) an operations plan, and (e) a training package.

(xvi) Prepare preliminary cost estimates for the recommended pilot cross-border facility.

(xvii) Analyze emerging social and environmental governance issues that stem from increased trade and transit between Pakistan and Afghanistan and their likely costs and other consequences. Develop data collection procedures and an analytical framework for monitoring such issues.

(xviii) Develop recommendations on preventing and mitigating of such costs and other adverse consequences.

(xix) Develop a manual with key elements of the social and environmental governance concerns and hold workshops in Chamman and in Islamabad and elsewhere to disseminate such manuals.

(xx) Assist in preparing environmental assessment including an environmental management plan for a follow-up subproject, supervise the implementation of mitigation measures, and train staff of CWD to address environmental problems related to CWD's operation.

C. Cost and Financing

3. The cost of the TA is estimated at $500,000. The TA will be financed by the TA funding program of the Asian Development Bank (ADB) on a grant basis. Detailed cost estimates are in the table.

Table A14: Cost Estimates

($'000)

Item	Units	Unite Rate	Foreign Exchange	Local Currency	Total Cost
A. Asian Development Bank Financing[a]					
1. Consultants					
a. Remuneration and Per Diem					
i. International Consultants	10	20	200	0	200
ii. Domestic Consultants	30	5		150	150
b. International Travel			5	0	5,000
c. Local Travel				10	10
2. Equipment			15	0	15
3. Vehicle Rental	20	15	0	30	30
4. Surveys			0	0	0
5. Communications and Reports				5	5
6. Workshop				10	10
7. Study Tour			20	0	20
8. Training			15		15
9. Training Materials			5		5
10. Representative for Contract Negotiations			5	0	5
11. Contingencies		285	20	10	30
Subtotal (A)			**285**	**215**	**500**
B. Government Financing					
Counterpart Staff, Office Space, Telecommunication			**0**	**130**	**130**
Total			**285**	**345**	**630**

[a] Financed by ADB's TA funding program.
Source: ADB staff estimates.

D. Implementation Arrangements

4. MOC will be the executing agency for the TA and will appoint a project director at joint secretary level. The established task force for trade facilitation at MOC, which is headed by the joint secretary (trade), will be actively involved during TA implementation, discussing and reviewing TA findings and providing guidance on various matters. The task force at MOC comprises members from the ministries of commerce, interior, and communication, National Highway Authority, National Logistics Cell, Central Board of Revenue, and from other public agencies, and representatives from the private sector.

5. The TA will be implemented 6 months and be completed in December 2004. The TA will require 10 person-months of international and 30 person-months of domestic consulting services by an (i) operations xxpert, (ii) legal experts, (iii) logistics expert, and (iv) social development expert. All consultants will be selected in accordance with ADB's *Guidelines on the Use of Consultants,* and other arrangements satisfactory to ADB for engaging domestic consultants. The quality- and cost-based selection procedure will be followed in selecting consulting firms. MOC has agreed to include relevant sections of ADB's *Anticorruption Policy* in all invitation bidding documents and contracts.

E. Reporting Requirements

6. The consultants will submit the following reports: (i) inception report, 3 weeks from the start of work in the field; (ii) interim report, within 3 months from start of the study; (iii) draft final reports, within 5 months and (iv) final report, 1 month after receiving comments from ADB and the Government. All reports will be saved and submitted in a compact disk.

SUMMARY ECONOMIC AND DISTRIBUTION ANALYSES

A. Methodology

1. The Balochistan Road Development Sector Project will involve improving some 1,400 kilometers (km) of selected national provincial and rural access roads. The economic analysis examined the expected contribution of the Project to the province of Balochistan. The two national highway sections and the six core roads[1] were subjected to detailed economic evaluation. Noncore roads will be subjected to a similar analysis for their inclusion in later stages of the Project. The Highway Development and Management Model of the World Bank (HDM-4) was used to analyze for direct savings in transport costs. Value added from ensuring new industries, employment, and sub-regional transport due to better transport links are not quantified. The main assumptions about maintenance and improvement strategies with and without the Project are discussed in paras. 2-4.

2. For the already paved national highways, the without project case is the situation where the quality of service provided by the existing roads is maintained at a reasonable level. The international roughness index (IRI) is allowed to deteriorate only until level IRI 10. At that point, periodic maintenance works are carried out to improve the condition to IRI 6. Routine maintenance activities that were considered use prevailing standards of the National Highway Authority: nominal patching of potholes, crack sealing by bituminous seal coat, grass cutting, grading of shoulders as well as cleaning of culverts and ditches, and refurbishing of sign posts. The maintenance strategy is assumed to remain constant for the 20-year evaluation period.

3. For provincial highways, the without project case is the situation where the quality of service provided by the existing roads is maintained at a reasonable level. IRI is allowed to deteriorate only until level IRI 10. At that point, periodic maintenance works are carried out to improve the condition to IRI 6. Routine maintenance activities and the overall maintenance strategy will be the same as in para. 2. For existing gravel/earth tracks, the roughness level of which is beyond measurement, only routine maintenance is provided. In recognition of the prevailing maintenance standards of Balochistan's Communication and Works Department, roughness levels are allowed to deteriorate up to the maximum value provided by HDM for such roads.

4. For the with project case, the two surface treatments were asphalt concrete (AC) and triple surface treatment (TST). Periodic maintenance based on road condition in terms of roughness level was assumed. Initial treatment will give IRI levels of 2.5 with AC and 4 with TST. For AC roads the maintenance strategy assumes that whenever IRI exceeds 6, an overlay of 50 millimeters (mm) for National highways or 25 mm for Provincial roads will be added with AC to lower IRI back to 2.5. For TST roads, the maintenance strategy assumes that whenever IRI exceeds 8, a 25 mm overlay will be added with TST to lower IRI back to 4. It is assumed that, after improvement, routine maintenance will be carried out: complete patching of potholes, regular ditch cleaning, refurbishing any safety measures disturbed, and grading of shoulders at regular intervals, especially after the monsoon.

B. Road Condition and Traffic

5. **Road Condition**. The condition of existing paved roads was investigated by topographic survey, pavement survey, roughness survey, and pit investigations. The results are summarized in Table A15.1. The of unpaved roads were visually inspected by the expert engineering staff.

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[1] The core roads were selected from a prescreened list of roads meeting the 12% minimum economic internal rate of return (EIRR) criterion of the Asian Development Bank.

Table A15.1: Road Condition Data

Road Section	Roughness Index (m/km)
National Highways	
Quetta – Kalat	8.44
Quetta - Chamman	8.42
Provincial Highway	
Sanjavi – Duki	11.47
Rural Access Road	
Dera Allah Yar to Hairdin	9.17

Source: .Consultants survey.

6. **Traffic Volume**. Traffic surveys were carried out on a normal day. A conservative factor of 1.0 was used to convert average daily traffic to average annual daily traffic based on local traffic patterns. The classified traffic count is shown in Table A15.2.

Table A15.2: Traffic Data for Project Roads, 2003

Road Section	Animal-Drawn	Motor-cycles	Cars	Wagons	Buses	Trucks 2 Axles	Trucks 3Axles	Trucks >3 Axles	Total Motorized Traffic
National Highways									
Quetta–Kalat	3	266	5,07	1,448	179	412	92	27	2,931
Quetta–Chamman	1	97	1,845	908	174	573	191	31	3,819
Provincial Highways									
Zhob–Mir Ali Khail–Khajuri Kuch	0	1	10	16	1	12	0	0	40
Nag–Punjgur	25	28	34	24	34	81	5	0	206
Kach–Shahrag–Harnai	0	15	25	25	8	265	15	0	353
Rural Access Roads									
Gulistan-Sagi Bazar–Kandel	8	91	52	221	1	7	0	0	372
Khurkhera–Kanraj	12	10	7	19	3	26	0	0	65
Dera Allahyar–Hairdin	145	228	201	290	31	197	16	2	965

Source: Consultants report.

7. **Traffic Growth Rate Assumptions**. The traffic growth rates (Table A15.3) were estimated following analysis of historical growth rates factors such as population, gross domestic product, value added in agriculture, transport and commercial fuel consumption, vehicle registration and various studies previously done by other institutions in Pakistan.

Table A15.3: Traffic Growth Rates

Category	Growth Rates
Car, Jeep, Taxi	5.0
Motorcycle	7.0
Wagon/Pickup	4.0
Bus	4.0
Truck	5.0
Multiaxle	5.0

Source: Consultants report.

C. Costs

8. **Vehicle Operating Costs**. For vehicle characteristics, default values of the HDM model were used. The other input data, namely, prices of vehicles and other inputs, and use of vehicles are in Table A15.4. The economic costs of representative vehicles, tires, and fuel were computed by deducting taxes and duties from market prices.

Table A15.4: Representative Vehicle Cost Data (PRs)

Vehicle	New Vehicle	Replacement Tyre	Fuel (per liter)	Crew (per hr)
Bus	2,455,800	5,391	16.40	50
Car	352,970	1,009	17.03	
Motorcycle	30,465	248	17.03	
2-Axle Truck	1,507,640	5,391	16.40	50
3-Axle Truck	2,059,130	5,391	16.40	50
> 3-Axle tuck	2,430,100	5,391	16.40	50
Wagon	1,161,850	2,130	16.40	50
Animal cart	200,000			20

Source: Government statistics and consultants's'calculations.

9. To have a realistic estimate of vehicle use, assumed values are shown in Table A15.5, using data available from other studies in Pakistan.

Table A15.5: Vehicle Use in Balochistan

Vehicle Category	Hours of Use per Year	Life (years)	Kilometers Covered per Year
Bus	1,750	10	70,000
Car	550	10	23,000
Motorcycle	400	10	10,000
2-Axle Truck	2,920	12	70,000
3-Axle Truck	2,920	12	70,000
> 3-Axle Truck	2,920	14	70,000
Wagon	1,500	10	60,000
Animal cart	1,300	3	4,000

Source: Government statistics and consultants calculations.

10. **Road Improvement and Maintenance Costs**. Improvement costs were established for each road separately on the basis of estimated quantities of work using the recently completed Rural Access Road Project in Balochistan. For national highways, old cost estimates of similar constructions were updated using current national highway rates. The costs based on the unit rates for various maintenance operations are in Table A15.6.

Table A15.6: Unit Maintenance and Operations Costs
(PRs)

Operation	Financial cost	Economic cost	Unit
Patching	156	133	m^2
Reseal	28	24	m^2
Surface dress TST	150	128	m^2
Overlay, 50mm	300	255	m^2
Shoulder edge Repair	40	34	m^2
Routine Maintenance	70,000	59,500	per km

TST= triple surface treatment.
Source: Asian Development Bank estimates.

11. The above financial costs, inclusive of taxes and duties, are converted into economic cost using the standard conversion factor of 0.9 as adopted in aq similar Asian Development Bank project in

Pakistan. It was assumed that the project duration was 2 years and/or 1 year for small roads, with 50% of the works to be carried out in year 1 and 50% in year 2 or 100% of the works completed in 1 year.

D. Benefits

12. The main quantifiable benefits of the investment are (i) the savings in Vehicle Operating Costs (VOCs) for existing traffic; and (ii) VOC savings to diverted traffic where applicable. A minor component will be savings in periodic maintenance cost to be incurred by the Government. The savings in VOC are due to better road surface conditions. As access to markets, which reflects agricultural benefits, improves, more economic activities are expected to result in generated traffic. However generated traffic of goods was ignored considering the water constraint on agricultural production, except for the Zhob Khajuri road, where current traffic is negligible and traffic levels have been predicted from adjacent land use data and land use patterns. The benefits to generated traffic are assumed to be half the rate for normal traffic.[2] The value of time used was 80% of the average wage for work journeys and 20% of the average wage for nonwork journeys.

E. Results of Economic Evaluation and Sensitivity Analysis

13. **Economic Evaluation**. Two improvement strategies, surface dressing with TST and AC was evaluated. For national highways, both AC and TST produced acceptable economic internal rate of returns (EIRRs) and can stand the worse case as in the sensitivity analysis. For the provincial highways and rural roads, both AC and TST proved economically acceptable except for the Khurkhera–Kanraj section that will only meet 12% criterion only by using TST. Based on the preceding analysis, AC and TST can be considered for national highways and core rural access and provincial roads, and TST is the preferred treatment for Khurkhera - Kanraj .

14. **Sensitivity Analysis.** Sensitivity analysis with TST was tested. Assumptions that would affect the outcome are listed in Table A15.7 and Table A15.8 along with a likely range of values. The analysis shows that all proposed roads tested still meet the 12% target with changes in the key assumptions over a significant range. Table A15.9 showed that the results are most sensitive to the assumed VOCs and capital costs. Respectively, a 20% reduction in VOC or an increase in capital costs of 17% is required for some of the core roads to be nonviable. While events of this magnitude are possible, the majority of the core roads require much larger changes, so the likelihood that the Project as a whole will prove nonviable is small. An extension in construction time by 1 year (delaying some costs and all the benefits) is equivalent to a 5% reduction in cost and a 12% reduction in benefits.

Table A15.7: Summary of EIRRs with Sensitivity Analysis for National Highways

Road Section	Strategy I (AC)	Strategy II (TST)	20% Reduction in VOC	50% Reduction in Value of Time	20% Increase in Capital Cost	Reduction in Growth from 5% to 3%	Extension of Construction Time by 1 Year
National Highways							
Quetta – Kalat	40.4	46.5	34.3	38.8	35.3	30.9	36.0
Quetta – Chamman	33.7	40.6	28.5	32.6	29.4	37.8	28.4

Source: Consultants' report.
AC = asphalt concrete, TST = triple surface treatment, VOC = vehicle operation cost

[2] This is the "rule of a half" normally used for estimating benefits to generated traffic.

Table A15.8: Summary of EIRRs with Sensitivity Analysis for Rural and Provincial Roads
(%)

Road Section	Strategy I (TST)	Strategy II (AC)	20 % Reduction in VOC	50 % Reduction in Value of Time	20% Increase in capital Cost	Reduction in Growth from 5% to 3%	Extension of Construction Time by 1 Year
Provincial Highways							
Zhob–Khajuri	17.5	13.8	14.0	16.2	14.7	16.8	14.6
Nag–Panjgur	28.6	23.9	25.3	27.4	24.4	26.6	24.1
Kach–Harnai	21.7	17.7	17.7	21.4	18.4	18.7	19.1
Rural Access Roads							
Gulistan–Kandel	48.9	41.5	40.8	45.0	41.3	46.6	34.1
Dera–Hairdin	27.3	23.6	24.8	26.3	24.3	25.3	23.8
Khurkhera–Kanraj	14.8	11.4	13.0	14.3	11.7	13.5	11.4

Source: Consultants' report.
AC = asphalt concrete, TST = triple surface treatment, VOC = vehicle operation cost.

Table A15.9: Switching Values (TST)

	20% Decrease in VOC		50% decrease in Value of Time		Growth Reduced from 5% to 3%		20% Capital Cost Increase	
Road Section	**NPV**	**SV %**	**NPV**	**SV %**	**NPV**	**SV %**	**NPV**	**SV %**
National Highways								
Quetta – Kalat	2,189	127	2,485	194	2,369	453	2,262	154
Quetta – Chamman	2,213	112	2,585	221	2,402	367	2,300	136
Provincial Roads								
Zhob–Khajuri	27	23	135	595	156	178	63	29
Nag–Panjgur	191	54	260	474	204	123	164	44
Kach–Harnai	434	78	571	302	480	226	462	96
Rural Access Roads								
Gulistan–Kandel	74	119	81	177	82	570	75	126
Dera – Hairdin	207	109	230	348	212	245	198	92
Khurkhera–Kanraj	7	30	18	610	9	68	(3)	17

SV Switching value– factor that will reduce the EIRR to 12%.
Source: Consultants' report.
NPV = net percent value, VOC = vehicle operating cost.

F. Distribution Analysis

15. Distribution analysis distributed all economic net benefits among the following categories of stakeholders: (i) local passengers, (ii) through travelers, (iii) freight users, (iv) vehicle owners, (v) unskilled labor, and (vi) Government/economy. In the case of Pakistan, both the passenger and freight transport markets are highly competitive. The proportion of the road user benefits that will be passed on to users was estimated to range from 70% for rural access roads to 90% for national highways. The majority of the vehicles are privately owned. For the Project's capital expenditure and maintenance cost on local labor, SWRF of 0.7 was assumed.

16. The summarized results of the distribution analysis are for national and provincial highways, and rural access roads.[3]

[3] The benefits do not include any estimate of the likely benefits of through traffic between Afghanistan and CAR and the Pakistan ports.

G. Poverty Impact

17. For the poverty impact analysis, assumptions were made for each road about the proportion of the poor in each stakeholder group. The incidence of poverty is very high in this region of Pakistan. Based on origin-destination surveys undertaken at 14 locations across Balochistan? where responses were correlated with income data obtained from the social survey? the distribution of the poor was developed for each road. The social survey revealed a high incidence of vehicle ownership amongst people below the poverty level.[4] Overall 20% of vehicle owners using the road were assumed to be poor (mostly motorcycle owners); 100% of unskilled workers were assumed to be poor, whereas skilled workers were assumed to be non-poor. Overall, the poverty impact ratio (PIR) is 0.46, higher than the 12.2% of the poor's share of national income or the head count poverty ratio of about 35% in the country. The value of the PIR is relatively insensitive to changes in the total benefits or costs.[5] The PIR is sensitive to the assumptions on the distribution of the poor among users. However, a 50% reduction in the proportion of the poor among local passengers would be needed to bring the ratio below 35%.

18. The overall result of distribution analysis is presented in TableA15.10

Table A15.10:Distribution of Benefits Analysis and Poverty Impact Ratio

Item		NET BENEFITS National Government Economic PV PRs (m)	Financial PV PRs (m)	Difference	DISTRIBUTION OF NET BENEFITS Road Users Local Passengers	Through Traffic	Freight Users	Vehicle owners	Labor	Government Economy
Inflows										
	Road User Benefits	20,419	0	20,419	7,571	3,299	7,114	2,337	0	97
Outflows										
	Initial Capital Investment	-6,628	-7,798	1,160	0	0	0	0	175	994
	Operation & Maintenance	-326	-390	64	0	0	0	0	35	29
Net Benefits		13,465	-8,188	21,653	7,571	3,299	7,114	2,337	210	1,120
Poverty Impact Ratio			Total							
	Gains and Losses			13,465	7,571	3,299	7,114	2,337	194	-7,068
	Proportion of the Poor (%)				40%	28%	35%	20%	100%	12%
	Net Benefits for the Poor			6,241	3,028	924	2,450	467	194	-862
		Poverty Impact Ratio		**0.46**						

Note: The net loss to Government/Economy (-7068) is calculated as –8188+1120.
PRs = Pakistan rupees, PV = present value,
Source: Consultants' report.

[4] This is believed to be due to the presence of smuggled or contraband goods, in which case, the actual use of vehicles is likely to be low.
[5] Except in the case of the Khajuri-Kanraj where the net benefit is small.

SUMMARY POVERTY REDUCTION AND SOCIAL STRATEGY

A. Linkages to the Country Poverty Analysis

Sector Identified as a National Priority in Country Poverty Analysis? Yes	Sector Identified as a National Priority in Country Poverty Partnership Agreement? Yes

Contribution of the sector/subsector to reduce poverty in Pakistan. Poverty in Pakistan has historically been higher in rural than in urban areas. Poverty rose more sharply in the rural areas in the 1990s, and in FY1999 the incidence of rural poverty (36.3%) was significantly higher than that of urban poverty (22.6%). During the 1990s, income distribution also worsened in rural areas, but improved slightly in urban areas. The increase in poverty is attributed mainly to declining economic growth, lack of social safety nets, persistence of severe macroeconomic imbalances, reduced flow of remittances from out-of-country Pakistani workers, and poor governance.

Poverty intervention is a critical theme of the Project. The direct and indirect potential for reducing poverty through road improvement will be realized through (i) increased access to markets, (ii) improved transportation of goods between areas of surplus and areas of deficit, (iii) reduced transport costs (potentially reduced cost of freight and goods as well as passenger fares), (iv) increased food security, and (v) increased employment opportunities. Improved transportation will improve the distribution of social services to the villages along the project roads. Local medical clinics and educational facilities will receive better support. Furthermore, road construction will create higher levels of income from agriculture and crop production; a demand for agricultural labor, wage rates, and wage income; and income from business and industries.

The Project will generate employment for people along the route. Unskilled laborers will be employed directly in road construction and maintenance and indirectly by providing material and services to the construction and maintenance activities.

More important are the long-term impacts of the Project. These are narrowing the gap of intradistrict disparities and providing the poor, unemployed youth, and women with better access to services and economic opportunities.

B. Poverty Analysis — Proposed Classification: PI

Based on the official poverty line set by the Government, 55% of the total population in the project area live below the poverty line (i.e., per capita income of less than PRs650 per month). In the latest report of the Social Policy and Development Center (SPDC), the 26 districts in Balochistan, Quetta, and Musa Khel rank as the least developed and most deprived.

A survey of the 53 proposed roads in 22 districts shows that (i) 55% of the population live below the poverty line, (ii) the education level in the districts is low (only 51% of males and 81% of females have primary education), (iii) 45% of the population are landless, and 58% do not own any land to cultivate.

During group interviews, perceptions of the Project were elicited from the local communities. The major findings are that the Project will (i) be beneficial as it will generate employment opportunities for the people, (ii) improve access to social services such as health and education, (iii) reduce transportation costs, (iv) improve their income as they can easily bring their goods to markets, (v) increase food security, and (vi) increase mobility of the population. The communities expect the Project to make a positive impact on their lives by raising the standard of living and reducing poverty in the province.

Poverty reduction will be monitored during and after construction to assess the impact of the Project on the poor. Benchmark poverty data (quantitative and qualitative) will be established prior to road construction. The institutional capacity of road Environment and Social Assessment Cell (RESAC) that will be established will be strengthened to address poverty, environment, social safeguard, and social development issues in the transport sector.

C. Participation Process

During project preparation, consultations were conducted with stakeholders in all project locations. The project beneficiaries are aware of the Project's long-term benefits and expressed their positive support for the Project. The Project will ensure that the affected people and various stakeholders are informed, consulted, and encouraged to participate actively for the duration of the Project. Nongovernment organizations (NGOs) will be involved in planning, implementing, and monitoring the resettlement plans.

D. Gender and Development

Women will be able to benefit from the Project through improved access to basic services such as health and education. Additional assistance to households with female heads will be provided as part of the resettlement plan.

Gender plan prepared? No

E. Potential Issues

	Significant/ Not Significant/ Uncertain/None	Strategy to Address Issues	Plan Required
Resettlement	Significant	A resettlement plan has been prepared for the two national highways and six core provincial roads. Land acquisition and resettlement are expected to be minimal, as few people live along these roads, road improvement will occur mainly along existing right-of-way and will involve mainly strip acquisition, where required. The summary of the resettlement plan is presented in Appendix 17. The resettlement framework (Appendix 18) will also be applied to the noncore subprojects.	Full
Indigenous People	None	Surveys conducted during project preparation revealed that the two ethnic groups (Baloches and Pathans) are all integrated into the mainstream life of the province. Therefore, a plan for development of indigenous people is not required.	Not applicable
Labor	Not significant	The Project will create employment opportunities and support local people to engage in road construction.	Addressed in the resettlement plan
Affordability	None	The development of the roads will reduce transport costs and increase the mobility of the local population in the project areas.	No
Other Risks/Vulnerabilities	Significant	Lack of capacity at the provincial level to address social and environmental safeguard issues will be targeted under the Project through capacity building and support. Vulnerable people will receive additional assistance as part of the resettlement plan.	Addressed in the resettlement plan

SUMMARY RESETTLEMENT PLAN

A. Background

1. The Balochistan Road Development Sector Project aims at widening and improving a total of 19 roads with an aggregate length of approximately 1,100 kilometers in Balochistan Province, Pakistan. The project roads are of three types: (i) national highways, (ii) provincial highways, and (iii) rural access roads. They include two national highway sections, namely Quetta–Kalat Road (N25, 130 km) and Quetta–Chamman Road (N25, 117 km) and six selected "core" provincial roads, comprising three provincial highways, namely: Kach–Shahrag–Harnai Road (95 km), Zhob–Mir Ali Khail–Khajuri Kach Road (101 km), and Nag–Punjgur Road (85 km); and, three rural access roads, namely: Gulistan–Sagi Bazar–Kandel Road (13 km), Dera Allahyar–Hairdin Road (40 km), and Khurkhera–Kanraj Road (24 km).

2. The Balochistan provincial government (BPG) selected the core provincial roads from roads prioritized by the project preparatory technical assistance consultants for (i) a geographic spread over different agro-ecological zones and major tribes (Pashtoon and Baloch) of the province; (ii) minimal adverse resettlement impact on the adjacent populations; (iii) poverty reduction; (iv) the likelihood of positive socioeconomic impacts on the province's population, particularly the poor and other vulnerable groups; (v) economic viability; (vi) intradistrict connectivity with markets and towns; and (vii) interdistrict, interprovincial, and international connectivity. The same selection criteria will serve largely as guiding principles for the remaining non-core roads under this sector Project.

3. The National Highway Authority (NHA) and the Communications and Works Department (CWD) will be the Executing Agencies (EAs) for their respective roads. Each EA will establish a Project Directorate in Quetta, which will have a Road Environment and Social Assessment Cell (RESAC) headed by a deputy director, an executive engineer-level officer to ensure proper planning, implementation, and monitoring of all land acquisition and resettlement-related activities. For noncore roads, each RESAC will conduct proper impact assessment, screen for possible land acquisition and resettlement effects and prepare a subproject Resettlement Plan (RP), full or short as required, for review and approval by the Asian Development Bank (ADB) before a civil works contract is issued.

4. This RP was prepared in accordance with ADB's *Policy on Involuntary Resettlement* and ADB's *Handbook on Resettlement*. The RP covers all eight core roads projects and presents the scope of land acquisition and resettlement, remedial measures for mitigating adverse impacts, and how and when these measures will be implemented. It is based on the census and socioeconomic assessment of the potentially affected persons (APs). Thus, this RP will serve as a model for RP preparation for other non-core subprojects. The preparation of an indigenous peoples development plan (IPDP), following ADB's *Policy on Indigenous Peoples,* is not required for this Project because the two ethnic groups (Baloches and Pathans) identified during the study are all integrated into the mainstream life of the province.

B. Scope of Land Acquisition and Resettlement

5. The proposed improvements will be carried out largely within the existing right-of-way (ROW) of the project roads. But the proposed widening will require additional 2- to 3- meter-wide strips of land on either side of most project roads. Over 90% of the lengths of the selected core project roads pass through nonproductive barren lands, which are known as statelands. For the remaining 5-10% lengths of the roads, the lands are owned by individuals who utilize them beneficially, i.e., for agricultural purposes in open areas and for residential and/or commercial purposes in built-up areas. These lands will be acquired as per provisions and rules

laid down in the Land Acquisition Act, 1894 (LAA, amended) in conjunction with ADB's resettlement policy and guidelines.

6. An estimated area of 42 hectares (ha) will need to be acquired for the two national highways (24 ha) and the six core project roads (18 ha). These are all privately owned productive and/or built-up areas, excluding the nonproductive tribal commons or statelands. The productive lands have mostly fruit orchards. Land acquisition will also affect an estimated 1,750 trees? 1,150 are fruit trees and 600 are wood trees (timber/firewood). As a result of additional land acquisition for the two national highways and six core provincial project roads, an estimated 344 people will be affected: 274 landowners and 70 tenants. Some realignments and bypasses may be necessary to minimize or avoid displacement in congested built-up areas and urban centers, which may be identified and decided upon later when the design is finalized. The realignments and bypasses (if any) will require the acquisition of additional landstrips as new ROW. New ROW will be selected carefully to minimize adverse resettlement and socioeconomic impacts. A complete inventory and valuation of the private lands to be acquired and the structures and other assets affected will be carried out and this RP will be revised accordingly.

7. A full census of all potentially affected households, small business enterprises (SBEs) and community/cultural structures along the project corridor of impact (the proposed formation width) was conducted on the eight selected roads in April-May 2003. The Project will partly affect 143 structures/assets, of which 80 are shops/SBEs, 39 are residential houses, 5 are cultural/community structures (mosques, etc.), 2 are enclosed fruit orchards, and the remaining 17 are other structures/asset, including 10 abandoned dilapidated structures. The Project will assist all structure owners to relocate on the available ROW land (out of the formation widths) so that APs can reestablish their businesses and restore their incomes with least disruption.

8. Land acquisition and removal of built-up structures affect 358 people on the provincial core project roads and the two national highways (247 km).

C. Objectives, Policy Framework, and Entitlements

9. In Pakistan, the Land Acquisition Act of 1894 (as amended) is the core legal tool the Government uses to acquire private lands for a public purpose. Under the provisions of the act, cash compensation for land is paid at the market rate to affected landowners, and for the affected crops only to registered tenants. The legal framework does not cover illegal occupants, squatters, and encroachers. In many project-specific cases, however, the governments of Pakistan and the other provinces (Sindh, Punjab and North West Frontier Province) have assisted project-affected persons even without any legal titles.

10. Pakistan has no specific policy on resettlement. In the absence of a formal policy to assist nontitled persons, a project-specific set of resettlement principles consistent with ADB policy requirements was adopted in this Project. As a result, APs, owners and tenants of acquired lands and affected structures (SBEs, houses, etc.), are entitled to a combination of compensation measures and resettlement assistance, depending on the nature of their ownership rights to lost/affected assets and scope of the impact, including social and economic vulnerability of the APs. In general terms, the APs will be compensated for the loss of lands and structures/assets, and assisted to restore their SBEs, livelihoods, and cultural/community structures/assets. Table A17.1 presents the details.

Table A17.1: Project Compensation Principles and Entitlements

Type of Losses	Entitlement
Barren nonproductive lands that are tribal commons/state Land	Free acquisition/utilization for the collective benefit of the local and far-off communities, as nobody will be adversely affected
Loss of agricultural land, crops and trees by owners (and sharecroppers)	Cash compensation for land based on current open market value, plus 15% compulsory acquisition surcharge (CAS), as provided in the Land Acquisition Act (LAA) Cash compensation for loss of crops at market value of mature crops Compensation to sharecropper/tenants as per the LAA Compensation for loss of wood trees at current market value Compensation for loss of fruit trees for average fruit production for the next 10 years to be computed at current market value Encroachers/squatters not eligible for compens ation for land
Loss of residential and commercial structures by owners	Genuine owners of the lands under built-up residential/commercial structures and similar other assets to be paid land compensation at current open market values plus 15% CAS, as provided in LAA Encroachers/squatters not be eligible for compensation for land Compensation for structures at full replacement costs Owners of affected structures to be allowed to take/reuse all salvageable materials for rebuilding/rehabilitating the structure Transfer allowance to cover the cost of moving structures (transport plus loading/unloading), including business premises Compensation for wells/handpumps/karez and other immovable assets at current replacement value, including installation charges
Loss of business premise by renters	One-time cash assistance equivalent to 2 months rent to renters to move alternative premise to reestablish their businesses Shifting allowance (transport plus loading/unloading charges) to be paid to affected businessmen (if applicable), on actual cost basis
Income assistance for loss of business by shops/SBE owners	One-time lump sum grant, based on the nature of business and type of losses; normally, the following range will apply: (i) small business, PRs.1,000-2,000; (ii) medium business, PRs.3,000-5,000; and large business/manufacturing, PRs.8,000-10, 000.
Loss of income by agricultural tenants/labor	One-time lump sum grant of PRs.1,000 to sharecrop/leaseholder tenants (in addition to their share in crop compensation). One-time lump sum grant of PRs.600 to regular/long-term agricultural labor (only those who are associated specifically with the affected land)
Loss of wages SBE employees	One-time financial assistance to hired labor equivalent to 30 days wages to be computed at local wage rates for various cadres Family workers in SBE not eligible, as they have already been covered by income assistance for loss of business of shops/SBE owners Special assistance of PRs.1,000 to affected vulnerable persons like the destitute, widows, and disabled to restore their livelihoods
Restoration of cultural/community structures	Complete rehabilitation/restoration by the Project; or, Cash compensation paid to the recognized patron/custodian for restoring affected cultural/community structures and/or installations

[a] If the replacement cost is higher than the amount of compensation, as assessed/determined by the district LAC, the difference will be paid by the Project (CWD/Balochistan or NHA/Quetta) in the form of cash assistance.

Source:

D. Consultation, Disclosure, and Grievance Redress

11. **Consultations**. The Project has strong support and positive public response as extensive community awareness and consultation programs among the affected/beneficiary communities were carried out during the field surveys. A total of 97 group interviews (men and women groups) were conducted along the core project roads. During project implementation, RESAC, assisted by the local nongovernment organization (NGO) and a resettlement specialist from the construction supervision consultants, will continue to prepare and implement the information and community consultation program in the project areas.

12. Disclosure of the RP. When the project design is finalized and the RP is subsequently revised, the Project will publish a project grief on its scope and objectives, including resettlement entitlements for APs, in both English and Urdu languages and distribute the same among the roadside communities, particularly the APs, by December 2003, at the latest. Generally, All literate people understand Urdu language. If necessary, however, the brochure will also be translated into Pashto, Brahvi, and Balochi languages. In addition, RESAC and the local NGO will hold discussion sessions with groups of APs to further explain and, if necessary, refine/improve the project brief.

13. **Grievances Redress Committee.** Provisions of LAA-1894 enable aggrieved APs to present their cases to the Land Acquisition Collector' office or to a court of law to have their grievances redressed and disputes resolved. The Project will establish a grievance redress committee (GRC), with representatives from the local government (union council, subdistrict/District nazims), local NGOs/CBOs, respective EAs (NHA and CWD) and APs. The GRC aims to provide a mechanism to mediate conflict and cut down on lengthy litigation, which often delays infrastructure projects.

14. **Resettlement Advisory Committee.** The resettlement advisory committee (RAC) will encourage local participation, ensure full transparency and accountability for the resettlement program and people's entitlements, and safeguard the rights of the most vulnerable peoples affected by the Project. The RAC will comprise representatives from the APs, NGO, CWD, and other stakeholders. Women will be represented on the RAC. The RAC will provide coordinating nodes for land acquisition and compensation, and relocation, and resettlement, and will assist the APs in all possible ways to minimize disruption and hardship during relocation.

E. Income Restoration Strategy

15. The entitlement matrix (Table A17.1) aims to compensate APs for their losses and to provide measures to minimize resettlement effects and restore or improve the APs's income/standard of living. Based on extensive consultations with the affected business owners and other stakeholders, affected SBEs will relocate on their own upon receipt of full compensation and before the start of civil works. The project directorates will notify their respective APs about 6 months ahead of the scheduled start of construction works. The APs will be allowed to take the salvageable materials for reconstructing their shops and will be assisted during relocation. The RESAC and the local NGO will assist the local business leaders and affected shop owners with better land use planning and will organize the relocated business area, taking into account traffic and safety of the road users.

F. Institutional Framework

16. Each EA will establish RESAC within its project directorate to implement the RP. Each EA will also engage a local NGO to help implement the RP, particularly for social preparation, disclosure of RP, public consultation, road safety awareness, and other resettlement-related activities, including grievance redress. A resettlement specialist from the project supervision consultants will also give technical assistance in resettlement planning, implementation, monitoring and evaluation.

G. Resettlement Budget and Financing

17. The total estimated cost for land acquisition and resettlement is expected to be in the amount of PRs.207.1 million ($3.6 million). The total RP implementation budget is estimated for 20 CWD roads included in the sector Project at PRs.159.1 million ($2.77 million), and for the two NHA roads at PRs.48.0 million (US$0.83 million). Balochistan's CWD will be spending PRs.44.5 million ($0.78 million) on RP implementation on the six core project roads, and the

remaining amount of PRs.114.3 million ($2.0 million) on 14 non-core subprojects. Both CWD and NHA at Quetta are committed to give their respective shares of the RP implementation budget. The detailed cost estimates are in Table A17.2.

Table A17.2: Detailed Cost Estimates for RP Implementation (in million)

	No.	RP Item	CWD Core	CWD Sector	NHA Sector	Total Sector Project PRs million	Total Sector Project $ million[a]
A.	1	Compensation for Additional Land Acquisition[b]	10.08	38.00	13.00	51.00	0.89
	2	Compensation for Affected Fruit/Wood Trees	9.00	36.00	12.00	48.00	0.83
	3	Compensation for Affected Structures/Assets	10.60	33.25	6.75	40.00	0.70
	4	Income Assistance to SBEs/Tenants	0.50	1.50	1.00	2.50	0.04
		Subtotal (A)	**30.18**	**108.75**	**32.75**	**141.50**	**2.46**
B.	5	Hiring of NGO to Assist in Implementing the RP	1.47	4.70	1.30	6.00	0.10
	6	Computer, Equipment and RESAC Staff Training	1.00	3.40	1.10	4.50	0.08
	7	Office, Transport for Main Office/Field Offices	1.00	3.40	1.10	4.50	0.08
	8	Monitoring Panel of Experts	0.88	2.75	0.85	3.60	0.06
	9	Short-Term Consultants (RP/Training/M&E)	0.88	2.75	0.85	3.60	0.06
		Subtotal (B)	**5.23**	**17.00**	**5.20**	**22.20**	**0.39**
		Total Program Cost (A+B)	**35.41**	**125.75**	**37.95**	**163.70**	**2.85**
	10	Administrative Costs (10% of A+B)	3.54	12.58	3.80	16.37	0.28
		Estimated Total Cost	**38.95**	**138.33**	**41.75**	**180.07**	**3.13**
	11	Contingencies (15% of Total Cost)	5.84	20.75	6.26	27.01	0.47
		Total (PRs. million)	**44.79**	**159.07**	**48.01**	**207.08**	**3.60**
		Total ($ million)	**0.78**	**2.77**	**0.83**	**3.60**	

[a] Exchange rate: PRs57.50 = $1.00.
[b] These amounts include 15% compulsory acquisition surcharge (CAS, as given in LAA, 1984.
Source: Consultants' report.
CWD = Communication and Works Department; M&E = monitoring and evaluation; NGO = nongovernment organization; NHA = National Highway Authority; RESAC = Road Environment and Social Assessment Cell; RP = Resettlement Plan; SBE = small business enterprise.

H. Implementation Schedule

18. NHA and CWD will implement the RP before civil works start in mid-/late 2004. A time-bound RP implementation schedule has been prepared in accordance with the time of civil works.[1] The Project will provide adequate advance notification to the APs and will pay compensation and resettlement assistance for lost assets at least 30 days before actual possession of the land or relocation of affected structures and assets. An identity card will be prepared for each AP and will be issued during the census survey.

I. Monitoring and Evaluation

19. RP implementation will be supervised and monitored by the project directorates, RACs, and GRCs. Monitoring will be both internal and external. The project directors (CWD and NHA) will prepare monthly reports on the progress of RP implementation. Project supervision consultants will monitor implementation and will report quarterly to NHA, CWD and ADB on the progress of resettlement activities. A panel of domestic experts, who will be engaged jointly by CWD and NHA, will conduct independent monitoring and evaluation twice a year for the first 2 years and then once every year during the remaining period of the Project. The independent monitoring and evaluation reports will be submitted to NHA, CWD, and ADB.

[1] A detailed time schedule for implementing the RP is given in the main RP document, and available upon request.

SUMMARY RESETTLEMENT FRAMEWORK

A. Purpose

1. The resettlement framework establishes the screening, planning, and management procedures for involuntary resettlement that apply to non-core road projects to conform to the policy on involuntary resettlement of the Asian Development Bank (ADB). The framework applies to activities prepared after Board approval and during loan implementation.

B. Procedures

2. The following procedures in preparing a resettlement plan (RP) for noncore road projects during project implementation will apply.

(i) Subprojects with possible land acquisition and resettlement effects are screened to ensure their consistency with ADB's policy on involuntary resettlement;
(ii) Resettlement effects are identified and avoided or minimized;
(iii) The need for a full or short RP is determined in accordance with the criteria set in ADB's *Handbook for Resettlement*,[1]
(iv) The project engineering cell prepares each subproject RP when required. The RP is submitted for approval to ADB and is implemented before the award of civil works contracts.

C. Scope

3. The resettlement framework has the following elements:

(i) **Scope of land acquisition and resettlement**. This covers (a) the scope and reasons for subproject resettlement, maps, any alternative options to minimize land acquisition, and main resettlement effects; (b) likely resettlement risks impoverishment or increased vulnerability for key stakeholders; and (c) primary responsibilities for resettlement.
(ii) **Socioeconomic information**. This comprise (a) census data, including numbers and spatial distribution of people affected by different types of loss; (b) methods for conducting socioeconomic surveys; (c) asset inventory; (d) likely resettlement effects; (e) all resettlement losses; (f) socioeconomic survey data of affected persons (APS) including household size, composition, and income and expenditure by key social groups; (g) definition of severely affected households; (h) impact on the poor and vulnerable groups (the destitute, widows, disabled, etc.) and any special measures needed for them, e.g., social preparation, special assistance in relocation.

D. Objectives, Policy Framework, and Entitlements.

4. In Pakistan, the Land Acquisition Act (LAA) of 1894 (as amended) is the core legal tool the Government uses to acquire private lands for a public purpose. Under the provisions of the act, cash compensation is paid at the market rate to affected landowners for land, and to registered tenants for the affected crops only. The legal framework does not cover squatters and encroachers. In many project-specific cases, however, the governments of Pakistan and

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[1] A full resettlement plan should be prepared if the subproject will have a significant impact on affected people. Resettlement will be “significant” where 200 or more people will experience major impacts. “Major” impacts constitute being physically displaced from housing and/or losing more than 10% of the household’s productive (income generating) assets.

the other provinces (Sindh, Punjab, and NWFP) have assisted even project-affected persons without any legal titles.

5. Pakistan has no specific policy on resettlement. In the absence of a formal policy to assist persons with no title to land, a project-specific set of resettlement principles consistent with ADB's policy requirements was adopted in this Project. As a result, the APs, the owners and tenants of acquired lands and affected structures (small business enterprises (SBEs), houses, etc.) are entitled to a combination of compensation measures and resettlement assistance, depending on the nature of their ownership rights to lost/affected assets and scope of the impact, and the social and economic vulnerability of the APs. In general terms, the APs will be compensated for loss of lands and structures/assets and assisted to restore their SBEs, livelihoods, and cultural/community structures/assets. Table A18 presents the details.

Table A18: Project Compensation Principles and Entitlements

Type of Loss	Entitlement
Loss of agricultural land, crops and trees by owners (and sharecroppers)	• Cash compensation for land based on current open market value, plus 15% compulsory acquisition surcharge (CAS), as provided in the Land Acquisition Act (LAA) • Cash compensation for loss of crops at market value of mature crops • Compensation to sharecropper/tenants as per LAA • Compensation for loss of wood trees at current market value • Compensation for loss of fruit trees for average fruit production for the next 10 years to be computed at the current market value • Encroachers/squatters not eligible for compensation for land
Loss of residential and commercial structures by owners	• Genuine owners of lands under built-up residential/commercial structures and similar other assets to be paid land compensation at current open market values plus 15% CAS, as provided in LAA • Encroachers/squatters not be eligible for land compensation • Compensation for structures at full replacement costs • Owners of affected structures allowed to take/reuse salvageable materials for rebuilding/rehabilitation of the structure • Transfer allowance to cover the cost of moving structures (transport plus loading/unloading), including business premises • Compensation for wells/hand pumps/karez and other immovable assets at current replacement value, including installation charges
Loss of business premise by renters	• One-time cash assistance equivalent to 2 months rent to the renters to move alternative premises for reestablishing businesses • Shifting allowance (transport plus loading/unloading charges) be paid to affected businessmen (if applicable), on actual cost basis
Income assistance for loss of business by shops/SBE owners	• One-time lump sum grant; based on the nature of business and type of losses. Normally, the following range will apply: (i) small business – PRs.1,000-2,000; (ii) medium business – PRs.3,000-5,000; and large business/manufacturing – PRs.8,000-10, 000.
Loss of income by agricultural tenants/labor	• One-time lump sum grant of PRs.1,000 to sharecrop/leaseholder tenants (in addition to their share in crop compensation). • One-time lump sum grant of PRs.600 to regular/long-term agricultural labor (only those who are associated specifically to the affected land)
Loss of wages by SBE employees	• One-time financial assistance to hired labor equivalent to 30 days wages to be computed at local wage rates for various cadres • Family workers in SBE not eligible, as they have already been covered by income assistance for loss of business of shops/SBE owners • Special assistance of PRs.1,000 to the affected vulnerable persons like the destitute, widows, and disabled for restoring their livelihoods
Restoration of cultural and community structures	• Complete rehabilitation/restoration by the Project (CWD/Bal'n.); or, • Cash compensation for restoring affected cultural/community structures and installations, to the recognized patron/custodian

[a] If the replacement cost is more than the amount of compensation, as assessed/determined by the District Land Acquisition Collector (LAC), then the difference is to be paid by the Project (CWD/Balochistan) in the form of cash assistance.

E. Relocation of Housing and Settlement

6. Relocation of the APs will be addressed as required in the subproject RPs through (i) options for relocating housing and other structures; site selection and planning; layout, and design; (ii) provision of basic social facilities; (iii) measures for integration with host communities; (iv) study of issues afftecting vulnerable groups; and (v) measures to manage and monitor any environmental risks.

F. Income Restoration Strategy

7. Income restoration measures will include (i) identifying livelihoods at risk, (ii) strategies for income restoration, (iii) plan for job creation, (iv) income substitution, (v) retraining, (vi) self-employment, and (vii) business relocation.

G. Institutional Framework

8. The project directorates of National Highway Authority (NHA) and Communication and Works Department (CWD) will execute the project resettlement activities. An executive engineer will be appointed as the deputy director of the Road Environment and Social Assessment Cell (DD/RESAC), in charge of land acquisition and resettlement operations, and will report to the project director. The DD/RESAC will plan, implement and monitor RP activities in close partnership with theAPs, affected/concerned district governments, land acquisition collectors (LACs), field offices, local nongovernment organization (NGO) hired to assist in RP implementation, resettlement specialist from the project supervision consultants, grievance redress committees (GRCs), and resettlement advisory committees (RACs). In addition, the RESAC will be responsible for periodic internal monitoring and evaluation of the project activities. A panel of experts will be engaged to evaluate RP implementation. Independent evaluations will be undertaken twice annually for the first 2 years and then annually during the remaining Project period.

H. Resettlement Budget and Financing

9. This will cover (i) measures for planning for land acquisition and budgeting and resettlement costs, annual budget, and sources of funding; (ii) arrangements for approval of RP cost estimates; and (iii) the flow of funds to reach APs and the contingency arrangements. Also included are guarantees to finance any additional costs required to meet the resettlement objectives, and confirmation of the delivery of entitlements to APs on time. NHA/CWD will ensure the prompt provision and efficient flow of the budgeted funds.

I. Implementation Schedule

10. An indicative schedule for RP preparation and implementation in relation to technical works will be prepared. It will show the time schedule of all resettlement activities, the start and finish dates before acquisition/demolition and procedures for implementing the key elements.

J. Monitoring and Evaluation

11. There will be arrangements for internal and independent monitoring and evaluation for RPs, including plans, timing, resources, and indicators for internal monitoring and external independent evaluation of resettlement targets; and an assessment of the capacity of internal and external monitoring and evaluation units.

ENVIRONMENTAL ASSESSMENT

A. Introduction

1. The Project is categorized as a B Project according to the environmental assessment guidelines of the Asian Development Bank (ADB). Therefore, the initial environmental examination (IEE) of the sample subprojects were prepared in accordance with ADB's environmental guidelines. This appendix is based on the IEE report of the sample projects and includes the assessment of the environmental impact on the sector.

B. Overview of the Environmental Assessment of the Sector Impacts

2. Balochistan Province's existing environmental problems are dominated by desertification, shortage of water due to lack of groundwater recharge, declining biodiversity, soil erosion and salination, and urban pollution especially water contamination. Although environmental protection and related natural resource management regulations such forest legislation exist in Pakistan, the present environmental administration in Balochistan is very weak. The Environmental Protection Ordinance of 1983 requires environmental assessment to be submitted to the provincial Environmental Protection Agency (EPA) for approval before any project is implemented. However, this practice was only recently implemented in Balochistan.

3. Improving the road network, as the main transport infrastructures in Balochistan will benefit the province by improving connectivity and facilitating traffic movement between rural areas and state national highways. However, it will also generate environmental costs related to road construction and operation. Increased mobility will lead to increased production and service sectors, and therefore to increasing deterioration of the environment and natural resources if appropriate mitigation measures are not applied. The environmental impacts associated with construction to improve the road networks will be addressed and handled mainly by the road authorities. However, dealing with the impact related to the operation of the road networks will involve other institutions and civil society. Better coordination among agencies including civil society is needed to anticipate and prevent environmental problems. Since the Project will strengthen the provincial Communication and Works Department (CWD) and establish the Road, Environment, and Social Assessment Cell (RESAC), it is expected that RESAC can facilitate coordination to address and handle environmental problems related to the transport sector in more sustainable and systematic ways. In addition, the technical assistance loan, which basically focuses on empowering civil society, will become a complementary component to maximize the project benefits and reduce any social and environmental costs related to the Project.

C. Summary of the IEE of the Sample Subproject

1. Project Description

4. The roads surveyed for the project covered 2,500 kilometers (km) in 21 districts of Balochistan. Subsequently, another 31 roads 508 km long in Kalat district were added. The National Highway Authority (NHA) and the government of Balochistan selected eight roads after initial prioritization: three rural access roads (RARs), three provincial highways (PHs), and two national highways (NHs). The RARs and PHs will involve construction or rehabilitation of the existing roads or tracks 3.7 meters wide and with a 2-meter shoulder on each side. The NHs will involve widening the existing roads to 7.3 meter. For all roads, culverts, bridges, and drainage will be rehabilitated.

2. Environmental Condition of the Project Area

5. The environmental conditions of the project area particularly along the road alignment are briefly summarized here.

6. **Gulistan-Sagi Bazar-Kandel Road (RAR)**. The present kutcha track of 13 km starts from Gulistan and ends at Kandel Road. It is flanked with fruit, vegetable, and wheat farms, irrigated mainly by tube wells. Water in this area has been pumped from a groundwater table at a depth of 300-600 feet. Farmers use the road to transport agricultural products to market. The present road crosses two small streams. The RAR does not pass through forest/game reserve, wildlife sanctuary, or any protected area. No endangered plant and animal species have been reported.

7. **Khurkhera–Kanraj Road (RAR)**. The present track extends to 24 km Lasbela district starting from 25 km after Khurkhera connection from NH25 and ending at village Pehi. The area is characterized by low-fertility gravel type soil on small bare mountains with low rainfall. There is no agriculture in the area and the inhabitants depend on livestock only. The area used to have fish and chicken farms, but these were ruined by the recent drought. Most of the land is devoid of any tree cover but sporadic bushes are found in patches. The road passes through flat drylands where no forest/game reserve, wildlife sanctuary, or any protected areas have been encountered. No endangered plant and animal species have been reported.

8. **Dera Allahyar–Hairdin Road (RAR).** This is 40 km road starts from Dera Allahyar near NH65) and ends at Hairdin. The road also leads to the Sui Gas fields. This is the only road in Balochistan that passes through fertile agricultural lands and has green cover along the road. The area is water rich due to the Put-Feeder canal. Limited industrial activity has been reported including brick kilns, cement pipe factory, rice mills, cotton ginning mills, and fish and poultry farms. More animal life is observed here due to the presence of water. The RAR passes through canal-irrigated lands, but no forest or game reserves, wildlife sanctuary, or protected areas have been encountered. No endangered plant and animal species have been reported.

9. **Zhob-Mir Ali Khail-Khajuri Kach Road (PH)**. This provincial single carriageway originates at Zhob (linking with NH50) and ends at Khajuri Kach. The road is located at the extreme north of the province connecting Balochistan to North West Frontiner Province via Tank. The land along the highway is highly mountainous with steep slopes and current pathway riding conditions are poor. Topography is undulating for the first 11 km while hilly areas occur for the rest of the road. The road meanders along Gomal, Zhob, and other rainwater streams, where isolated and protruding rocky mountains are located. There is a katcha (unpaved) shingle track to Mughal Kot, 80 km from Zhob, and another 20 km track that cannot sustain any type of traffic. Grapes and apricot are major fruits being grown. Forest cover is rare in the area, except greenery and agro-based activity along the river where water flows even during drought. In rainy season, all water channels are full, disrupting the transport route and emergency supplies to the local people. The groundwater table is in the range of 50-100 feet in and around Zhob. Agroforestry along water channels has provided alternate habitats for many birds, insects etc. A few unexplored coal mines at the end of the road may be exploited on completion of the road. In contrast with that of other selected roads, this area is relatively less polluted by dust due to better vegetative cover. Most of the corridor along this road consists of gravel-type mountains with possibility of some landslides. The highway does not pass through forest or game reserve, wildlife sanctuary, or any protected area. No endangered species have been reported.

10. **Nag–Punjgur Road (PH).** This PH originates at the outskirts of Panjgur and ends at Nag (Kharan and Panjgur districts). Current PH riding conditions are not good and the PH is unpaved (katcha track with loose gravel top). Fruit farms are observed only in Panjgur area. Beyond Panjgur, very little habitation is possible because drought conditions, sandy soils, and hardened surfaces make it unfit for vegetation and fruit farming. Again the water table is reported to be lower than in other parts of Balochistan, i.e., 25-300 feet deep but is not extracted because financial resources of the inhabitants are low. Wind is reportedly strong most of the year along the road and can be used to generate electricity, pumping water, etc. Population density along the roads is low. The road runs almost on level ground with sandy soil and gravel-black mountains devoid of any bushes or trees, adjoining vast stretches of uncultivated dry field. The area has very little green cover or plantations as a whole and faces sandstorms regularly resulting in visibility problems. The road crosses a number of dried-up streams and rivers including Rakhshan River, Surab River, and Sabzab Nullah. No forest or game reserves, wildlife sanctuary, protected areas, or endangered species have been reported.

11. **Kach–Shahrag–Harnai Road (PH).** This unpaved provincial road starts from Kach located 71 km from Quetta on Quetta-Ziarat Road and ends at Harnai. It covers two districts of Ziarat and Sibi. There is an old outdated railway tunnel through which the current road passes, constructed to avoid land sliding onto the railway tracks. A few pockets of loose soil were observed, which makes landslides and erosion of the embankment common. Pockets close to the riverbed show scattered fruit and vegetable farms and medicinal herbs. Some parts of the valley along the road depend on surface rainwater flowing through the rivers and is called barani-fed areas, with groundwater table as low as 200 feet. Other areas have groundwater at a depth of 500-800 feet. During the heavy monsoon season flooding closes portions of the road. The highway crosses the river Khost, which had water flowing throughout the year before the dry spell of the past 5-6 years, requiring a number of bridges on the river and eroding bases of the existing road. This road is used to transport fruits/vegetables to markets, coal to nearby thermal power plants and other users in Balochistan and Punjab; and to bring wood from Punjab. Coal mining is common in Harnai, which has four railway stations involved in coal transport throughout the country. The railway line runs from Khost to Harnai and beyond. The highway passes through partially irrigated lands, but there is no forest or game reserve, wildlife sanctuary, or any protected areas. No endangered species have been reported.

12. **Quetta-Kalat Road (NH)**. This section of the road starts from the outskirts of Quetta and ends at Kalat. This road is an important part of NH25 originating from Karachi to Chamman and would be the key route to Afghanistan and further north to the Central Asian republics. This highway passes through partially irrigated lands with fruit and agricultural farms. Soil is patchy, with gravel on most of the route, but sandy soil occurs after Pringabad with typical desert bushes in the area. The land is fertile but lacks water resources and supply and depends heavily on either pumping of underground water or rainwater harvesting. The water table depth varies from 350 to 600 feet in towns along the highway. The road crosses a number of small towns and markets like Pringabad, Mastung, Mangochar, and Kalat. At Lakh pass, an 80-m mountain cut is being established to minimize gradient and turning. Heavy winds and dust storms are typical in certain areas, especially after Pringabad. Mixed types of trees are planted near Mastung. This area is designated as an earthquake zone. Its history of earthquakes in the recent past dictates the architecture of homes which are constructed with mud to minimize losses and blend with nature. The project road is 5 km away from the Hazarghanji Park. No other forest or game reserve, wildlife sanctuary, or any protected areas have been encountered. No endangered species have been reported.

13. **Quetta–Chamman Road (NH)**. The road starts from the outskirts of Quetta and ends at Chamman. It forms part of the integrated program for improvement and upgrading of NH25 from Karachi to Chamman. All rivers and streams have dried up. Land in the corridor consists of loose soil without any tree cover. In windy weather, dust storms are common. The few inhabitants of the area depend on

livestock. Close to Quetta, a few stone crushers operate. Brick kilns using local coal with high emissions of visible smoke are scattered along the road to Chamman. The road passes through a number of towns: Beleli, Kuchlak, Yaru, Saranan, Jungle Pir Ali Zai, Killa Abdullah, ShelaBagh, Khojak Pass, and Chamman. Travel time between Quetta and Chamman is around 2 hours 20 minutes with the existing condition of the road. A small dried-up river/stream winds along the road after Killa Abdulllah for 20-30 km and the existing road alignment crosses a number of rainwater streams. Kojak mountain terrain near Shelabagh is a difficult terrain with high water-induced landslides and erodible soil. The PH passes through scantily irrigated lands where no forest or game reserve, wildlife sanctuary, or any protected areas have been encountered. No endangered species have been reported.

3. Screening Potential Impact and Mitigation Measures

14. No significant environmental impact is expected due to location because the Project will be confined to rehabilitating existing roads and does not include roads located nearby or within reserve areas or wildlife sanctuaries. Environmental impacts may include (i) disrupted traffic in the construction area, (ii) increased air pollution due to increased dust and other volatile chemical substances from the asphalt plant, (iii) noise and vibration, (iv) disrupted water systems due to cut and fill and other earthwork, and (v) potential landslides associated with elevating the road. Such impacts will occur only during the construction period. Mitigation measures will include (i) rerouting traffic during the construction stage; (ii) maintaining optimum moisture content during handling of soil, spraying water to minimize dust, and maintaining a safe distance between asphalt plants and public facilities including schools; (iii) strictly controlling the construction works that create noise and vibration by prohibiting night work near residential areas; (iv) pumping stagnant water and providing adequate drainage systems; (v) stabilizing road embankment sideslopes; (vi) avoiding road widening at ponds and widening roads only on one side to avoid excessive tree cutting; (vii) planting trees; and (viii) establishing forums to communicate with affected persons. All these mitigation measures will be included in the contract documents for the contractor. All the impacts are clearly temporary and manageable. However, special attention to implementing environmental mitigation is needed for the following:(i) roads connecting Zhob to Khajuri and Quetta-Chamman due to the need for heavy cutting of mountains and landslide problems; and (ii) road connecting Dera Allahyar to Hairdin due to potential accidents related to the gas pipeline. Improving the road will reduce dust, but it will also increase traffic, which might increase air pollution related to motor vehicle operation. Since the cause of the impacts are beyond the responsibility of the EA, it is necessary that the EA maintain tree planting along the road to minimize the impact of vehicle emission on public health and to maintain the embankment to avoid erosion. Borrow pit operations will be strictly controlled (e.g., no borrow pits will be allowed in agriculture lands or areas prone to erosion and landslide, and former borrow pit areas will be rehabilitated). Consultations carried out with communities in the surrounding project areas during the IEE noted that the people expected the Project to benefit them by providing quicker assistance for medical patients, minimum fares, and shorter time for their agricultural produce to reach the market.

4. Institutional Arrangement for Monitoring and Implementing Environmental Management Plan

15. The environmental management plan (EMP) covers actions to ensure that mitigation measures and monitoring of theie implementation throughout project implementation and operation are part of the IEE. The EMP stated that, during construction impacts associated with earthwork (e.g., siltation, erosion, dust, noise) need to be carefully mitigated. During operation, air pollution due to vehicle emissions and contamination of oil are impacts that need to be handled. Implementation of the mitigation measures will be monitored in accordance with the Government and ADB's environmental assessment requirements. The report on EMP implementation should be submitted to ADB annually. Based on the EMP,

implementating mitigation measures during the construction stage will be the responsibility of the contractor.

16. An environmental specialist will be hired as part of the consultant team for implementing the Project and will assist the RESAC in Balochistan CWD to prepare contractual documents so that the bid documents and other obligations of the contractors clearly identify their environmental responsibilities and describe penalties for noncompliance. In addition, the environmental specialist as part of the consultant's team will also provide recommendations to the engineering design team to address environmental concerns, and supervise monitoring of mitigation measures implementation during the construction stage. To deal with environmental impacts associated with the operation, particularly with increasing vehicle emission, oil contamination due to spillage and linkage, Balochistan CWD will strengthen coordination with relevant agencies to minimize the impacts. CWD and its RESAC will implement overall environmental monitoring and management, while NHA will implement its EMP.

5. Finding and Recommendations

17. The present IEE reveals that few environmental impacts are likely to occur due to the construction activities; normal operations of the proposed rehabilitation project and recommendations are suggested accordingly for mitigating these perceived impacts. A full environmental impact assessment (EIA) for roads involved in this Project is not required.

D. Environmental Criteria for Subproject Selection

18. The following selection criteria will be adopted for the follow on subprojects:

(i) The subproject will not involve any widening of rights-of-way of the existing roads located within the areas declared as environmentally and socially sensitive (e.g., national parks, sanctuaries, cultural heritage).
(ii) The Project will involve only the subprojects that comply with all necessary Government environmental requirements such as a no objection certification from Pakistan or Balochistan EPA.

E. Review Procedures for the Environmental Assessment of the Follow-on Subprojects

19. The EIA for the follow-up subproject will be prepared and reviewed by the following procedures:

(i) The IEE (including the EMP) for all follow-on subprojects must be prepared by the environmental specialist as part of the consultant team. If any significant environmental impact is predicted, an EIA will also be prepared.
(ii) During the IEE/EIA, adequate consultations with affected groups will be carried out and a report on these consultations will be an appendix of the IEE/EIA.
(iii) All the IEE reports will be reviewed by RESAC. All the IEE reports will be submitted to ADB and the Pakistan or Balochistan EPA. The construction works can start only after ADB concurs with the IEE report and the no objection certificate granted by the Pakistan or Balochistan EPA.